DEVELOPMENT OF AN OPTICAL TIME-DOMAIN REFLECTOMETER-BASED TEMPERATURE SENSOR EMPLOYING A D-SHAPED FIBER

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by

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TABLE OF CONTENTS

ACKNOWLEDGEMENT..... ii
TABLE OF CONTENTS.....iii
LIST OF TABLES..... v
LIST OF FIGURES.....vi
LIST OF SYMBOLS..... vii
LIST OF ABBREVIATIONS..... ix
ABSTRAK..... x
ABSTRACT..... xii
Chapter 1: Introduction.....1
1.1 Background.....1
1.2 Problem Statement.....2
1.3 Aim and Objectives.....3
1.4 Scope of the Study.....4
1.5 Significance of the Study.....4
1.6 Thesis Organization.....5
Chapter 2: Literature Review.....6
2.1 Introduction.....6
2.2 Temperature Sensor Based on Optical Fiber.....7
2.2.1 Wavelength-Modulated Temperature Sensor.....7
2.2.2 Phase-Modulated Temperature Sensor.....9
2.2.3 Attenuation-Modulated Temperature Sensor.....11
2.3 Principles and Applications of Optical Time-Domain Reflectometer (OTDR).....14
2.3.1 OTDR Operating Principles.....14
2.3.2 Fresnel Reflection-Based Sensing Technique.....16
2.3.3 Previous Works on OTDR-Based Temperature Sensing.....19
2.4 Physics of Optical Fiber.....22
2.4.1 Propagation of Light in Optical Fiber.....22
2.4.2 Thermo-Optic Effects in Silica Fiber.....24
2.5 Fabrication and Sensing Principle of D-Shaped Fiber.....26
2.5.1 Fabrication and Geometry of D-Shaped Fiber.....26
2.5.2 Evanescent Field and Modified Lambert-Beer Law.....28
2.5.3 Previous Works on Temperature Sensor Based on D-Shaped Fiber..... 31
Chapter 3: Methodology and Experimental Setup.....34
3.1 Experimental Overview and System Architecture.....34
3.2 Fabrication of Sensing Probe.....37
3.2.1 Fiber Preparation.....37

3.2.2 D-Shape Polishing.....40
3.2.3 Geometric Characterization of the Sensing Probes.....42
3.3 Experimental Setup for Temperature Sensing.....43
3.3.1 OTDR Configuration and Parameter Selection.....43
3.3.2 Heating Element and Thermal Control.....46
3.3.3 Data Acquisition Process.....46
Chapter 4: Results and Discussion.....50
4.1 Sensing Probes Geometric Characterization.....50
4.2 Temperature Sensing Performance of Bare Fiber.....51
4.3 Temperature Sensing Performance of D-Shaped Fiber.....54
4.4 Thermal Stability of Sensing Probes.....58
Chapter 5: Conclusion.....61
5.1 Summary..... 61
5.2 Future Work and Recommendations.....62
REFERENCES..... 64

LIST OF TABLES

Table 2.1 Comparison of D-shaped fiber-based temperature sensors reported in the literature.

Table 3.1 OTDR measurement parameters

Table 4.1 Comparative sensitivity and linearity across interrogation pulse widths

Table 4.2 Statistical summary of peak signal level stability over 10 consecutive trials at 60℃.

LIST OF FIGURES

Fig. 2.1 Typical OTDR test curve showing Rayleigh scattering and Fresnel reflection peaks.

Fig. 2.2 OTDR-based sensor for temperature monitoring: (a) SMS (Singlemode-Multimode-Singlemode) fiber structure and (b) schematic setup for measuring temperature variation using an OTDR.

Fig. 2.3 Total Internal Reflection (TIR) illustrated in (a) glass and air mediums and (b) a fiber structure with different refractive indices.

Fig. 2.4 Schematic diagram of a D-shaped fiber showing its asymmetric geometry.

Fig. 3.1 Flowchart of Research Methodology.

Fig. 3.2 Measurement setup for the characterization of the D-shaped fiber temperature sensor, including the OTDR, fiber spool, and reference thermocouple.

Fig. 3.3 Fabrication of bare fiber: (a) the cleaving process and (b) the fusion splicing process using a Signal Fire AI-9.

Fig. 3.4 Custom setup used to fabricate D-shaped fiber, featuring a rotary polishing tool and real-time power monitoring.

Fig. 4.1 Side-view optical microscope images of the fabricated probes: (a) bare fiber, and (b) 2 dB, (c) 4 dB, (d) 6 dB, (e) 7 dB, (f) 8 dB D-shaped sensing probes with increasing polishing depth. Digital profiling lines are relative geometric approximation across the sample set.

Fig. 4.2 OTDR peak loss as a function of temperature for the reference bare fiber at (a) 5 ns, and (b) 10 ns interrogation pulse widths

Fig. 4.3 Combined temperature response curves and linear regression fits for the 2 dB, 4 dB, 6 dB, 7 dB, and 8 dB D-shaped fiber probes interrogated at a 5 ns pulse width.

Fig. 4.4 OTDR trace at (a) the 2 km marker and (b) zoomed peak level against temperature variations for the 7 dB D-shaped sensing probe across a temperature range of 40℃ to 80℃ at 5 ns pulse width.

Fig. 4.5 Peak signal level stability curves and linear regression fits over consecutive trials at a static temperature of 60°C for (a) the reference bare fiber control probe and (b) the 7 dB D-shaped sensing probe.

LIST OF SYMBOLS

λ_B	Bragg wavelength
n_{eff}	Effective refractive index of the fiber core
n_{gr}	Group index of the glass core
Λ	Grating Period
$P_S(L)$	Backscattered optical power received by the OTDR detector at a given distance L
L	Distance along the optical fiber / total length
S	Backscatter capture fraction / sensitivity coefficient
α_s	Rayleigh scattering coefficient
W	Spatial length of the propagating optical pulse
P_0	Initial peak power of the injected laser pulse
a	Total intrinsic attenuation coefficient of the fiber / absorption coefficient
τ	Temporal pulse duration (laser pulse width)
v_{gr}	Group velocity of the guided light wave
c	Speed of light in a vacuum
Δz	Spatial resolution of the OTDR instrument
R	Fresnel reflectance fraction
n_f	Refractive index of the fiber core
n_m	Refractive index of the surrounding external medium
T	Temperature value
$\partial n/\partial T$	Thermo-optic coefficient of a material
α_n	First-order linear thermo-optic coefficient
β_n	Second-order quadratic thermo-optic coefficient
C_0	Constant offset value
n_{co}	Refractive index of the waveguide core matrix
n_{cl}	Refractive index of the surrounding cladding layer
θ_1	Angle of incidence at the interface boundary
θ_2	Angle of refraction across the boundary
θ_c	Critical angle required for total internal refraction
n_0	Refractive index of air
θ_a	Maximum acceptable coupling angle in air
V	Normalized frequency parameter
a	Physical core diameter of the optical fiber
λ_0	Operational light propagation wavelength
$E(x)$	Amplitude of the decaying electric field at a transverse distance x
E_0	Maximum electric field magnitude evaluated exactly at the core interface

	boundary
x	Transverse distance extending away from the core-cladding boundary line
d_p	Penetration depth of the evanescent field
h	Polishing depth
D	Residual cladding thickness
A	Optical absorbance magnitude
η	Fractional power of evanescent field
ϵ	Molar absorptivity of a chemical species
C	Concentration level of a target molecular species

LIST OF ABBREVIATIONS

ALD	Atomic Layer Deposition
AWG	Arrayed Waveguide Grating
BOTDA	Brillouin Optical Time-Domain Analysis
BOTDR	Brillouin Optical Time-Domain Reflectometer
CW	Continuous-Wave
DI	Deionized
DTS	Distributed Temperature Sensors
EMI	Electromagnetic Interference
FBG	Fiber Bragg Grating
FOEW	Fiber Optic Evanescent Wave
FPI	Fabry-Perot Interferometer
MMF	Multi-mode Fiber
MZI	Mach-Zehnder Interferometer
NA	Numerical Aperture
OFDR	Optical Frequency-Domain Reflectometer
OFS	Optical Fiber Sensors
OLS	Ordinary Least Squares
OTDR	Optical Time-Domain Reflectometer
PCF	Photonic Crystal Fiber
PR-LPG	Photoresist Long-Period Grating
PW	Pulse Width
RI	Refractive Index
ROTDR	Raman Optical Time-Domain Reflectometer
RTD	Resistance Temperature Detector
SMF	Single-mode Fiber
SMS	Singlemode-Multimode-Singlemode
SNR	Signal-to-Noise Ratio
SPR	Surface Plasmon Resonance
TIR	Total Internal Reflection
UV	Ultraviolet
VFL	Visual Fault Locator
WDM	Wavelength-Division Multiplexing

PEMBANGUNAN SENSOR SUHU BERASASKAN REFLEKTOMETER DOMAIN MASA OPTIK MENGGUNAKAN GENTIAN BERBENTUK D

ABSTRAK

Aplikasi Reflektometri Domain Masa Optik (OTDR) untuk sensor suhu biasanya dihadkan oleh sensitiviti yang lemah dan nisbah isyarat-ke-hinggar (SNR) yang rendah akibat sifat lengai gentian optik terhadap haba persekitaran. Bagi mengatasi halangan ini, satu pengubahsuaian struktur gentian berbentuk D yang dipotong telah dicadangkan. Kajian ini memfokuskan kepada pembangunan prob gentian berbentuk D melalui teknik pengilapan sisi yang dicirikan oleh kepelbagaian kedalaman pengilapan, serta penilaian terhadap prestasi pengesanan suhu dengan OTDR sebagai instrumen pengesanan. Secara khususnya, lima konfigurasi prob gentian berbentuk D yang berbeza kehilangan penyisipan awal semasa proses pengilapan in-situ telah dihasilkan: 2 dB, 4 dB, 6 dB, 7 dB, dan 8 dB, di samping prob gentian yang tidak digilap sebagai rujukan kawalan. Prob gentian berbentuk D tersebut direndam di dalam bikar mengandungi air dinyahion (DI) yang diletakkan di atas plat pemanas dan plot OTDR direkodkan dalam julat suhu 40 ℃ hingga 80 ℃ dengan kenaikan suhu sebanyak 5 ℃ dengan lebar denyut 5 ns dan 10 ns. Hasil kajian menunjukkan bahawa prob gentian yang tidak digilap menunjukkan sensitiviti pada 0.0006 dB/℃ dengan R^2=0.8259, manakala lebar denyut 10 ns mempunyai sensitivity sebanyak 0.0001 dB/℃ dan R^2=0.8048. Fabrikasi prob gentian berbentuk D menunjukkan peningkatan prestasi dengan prob gentian berbentuk D 7 dB dengan lebar denyut 5 ns mencatatkan nilai lineariti tertinggi (R^2 = 0.9552) dengan sensitiviti sederhana pada 0.0008 dB/℃, namun prestasi merosot bagi prob gentian berbentuk D

8 dB (R^2 = 0.4422). Ujian kestabilan yang dijalankan pada prob gentian berbentuk D 7 dB pada suhu 60 ℃ selama 20 minit menunjukkan prestasi yang agak stabil dalam persekitaran terma statik, dengan sisihan piawai 0.00236 dB. Hasil kajian ini membuktikan potensi pembangunan OFS suhu berstruktur mudah dengan kebolehpercayaan tinggi bagi tunjuan pemantauan suhu masa nyata.

DEVELOPMENT OF AN OPTICAL TIME-DOMAIN REFLECTOMETER-BASED TEMPERATURE SENSOR EMPLOYING A D-SHAPED FIBER

ABSTRACT

The application of Optical Time-Domain Reflectometry (OTDR) for temperature sensor comes at a cost of weak sensitivity and low signal-to-noise (SNR) ratio due to the inertness of optical fiber to the thermal environment. A structural modification of cleaved D-shaped fiber is proposed to overcome this limitation. This study focus on the development of D-shaped fiber probe via the side-polishing technique, characterized by variants of polishing depths, and investigation of the temperature sensing performance with OTDR as the interrogation instrument. In particular, five D-shaped fiber probe configurations with different initial insertion loss during the in-situ polishing process are fabricated: 2 dB, 4 dB, 6 dB, 7 dB, and 8 dB, and a bare fiber as a controlled reference is prepared. The D-shaped fiber probe is submerged in a beaker of deionized (DI) water placed on top of a hot plate and OTDR traces are recorded over a temperature range of 40 ℃ to 80℃ with 5℃ increment with pulse width (PW) of 5 ns and 10 ns. The results show that the bare fiber probe with 5 ns PW has a sensitivity of 0.0006 dB/℃ and R^2=0.8259, while 10 ns PW with sensitivity of 0.0001 dB/ ℃ and R^2=0.8048. The fabrication of D-shaped fiber probe improve the sensing performance with the optimal 7dB D-shaped fiber probe recorded on 5 ns PW has the highest linearity (R^2 = 0.9552) with moderate sensitivity of 0.0008 dB/℃, but the performance degraded for the 8 dB D-shaped fiber probe (R^2 =0.4422). The stability test conducted on the 7 dB D-shaped fiber probe at 60 ℃ over a 20-minute period indicate a fairly stable performance in a static thermal

environment, yielding a standard deviation of 0.00236 dB. The result of this study demonstrates the possibility of developing a structurally simple temperature OFS with a high reliability for real-time thermal monitoring.

Chapter 1: Introduction

1.1 Background

Temperature is one of the most fundamental and widely measured physical parameters in both science and industry. There is a critical demand for accurate and reliable temperature monitoring across wide range of applications, such as structural health monitoring, thermal management of power infrastructure, oil and gas pipeline inspection, industrial process control, medical diagnostics, and environmental monitoring. Research in developing a more robust and sensitive temperature sensing technology remains active and continue to grow as engineering environments become more complex.

Traditional electronic temperature sensors, such as thermocouples, resistance detectors (RTDs), and thermistors, have served as industry standard for decades. However, they come with several limitations that restrict their utility in certain critical environments. Electrical-based sensors are inherently susceptible to electromagnetic interference (EMI), posing reliability concerns in high-voltage or electrically noisy settings, and present safety risks in chemically hazardous or explosive environments. Optical fiber sensors (OFS) directly address these shortcomings. Since the sensing medium is optical rather than electrical, OFS are inherently immune to EMI, chemically inert, and lightweight [1]. These properties make optical fiber a compelling platform for temperature sensing in environments where conventional electronic sensors are inadequate.

Among the interrogation techniques available for OFS, Optical Time-Domain Reflectometer (OTDR) is one of the most practical and cost-effective option. An

OTDR operates by launching a short optical pulse into one end of a fiber and recording the backscattered light returning to the same port as a function of time, which is converted into distance along the fiber. Any localized perturbation at a sensing element, such as temperature-induced change in optical loss, appears as a measurable event on the OTDR trace at the corresponding length of the fiber [1]. This single-ended operation is a practical advantage, as it requires access to only one end of the fiber. Furthermore, commercial OTDR instruments are widely available and cost-effective, making them an attractive interrogation platform for point sensing applications where a sensing element is placed at a defined position along the fiber.

To make an OTDR-based sensor responsive to temperatures, the sensing element must be designed to translate temperature changes into a measurable optical loss. A D-shaped optical fiber, which is a standard single-mode fiber with one side of its cladding mechanically removed through controlled polishing, is ideal for this. This asymmetric geometry reduces the distance between the fiber core and the external environment to just a few micrometers, allowing the evanescent field of the guided light to interact with whatever medium or material placed on the polished surface [2]. It is also possible to further increase the temperature-dependent loss detected by OTDR by applying a thermally responsive coating that will alter the optical properties of the sensing interface on which it is deposited on [3].

1.2 Problem Statement

A wide range of optical fiber temperature sensors utilizing variety of architectures have been developed and reported in the literature. One such architecture with high potential is OTDR as the main interrogation system. However, comparing to other well established temperature OFS, such as Fiber Bragg Grating (FBG) and

Fabry-Perot interferometer (FPI) sensors, the main limitation of OTDR system is in terms of its performance due to being fundamentally restricted by the optical sensitivity of the sensing element itself. When operating along a standard glass fiber, the temperature dependence is too weak that it is ineffective to function as a sensor due to the limited interaction between the guided optical field and the external environment, hence decreasing the sensitivity of the sensor [4].

To overcome this limitation, a structural modification is proposed, in particular D-shaped geometry modification fabricated via side-polishing technique. Several studies have shown this particular fiber structure is capable of enhancing the performance of temperature sensing by exposing the leaked evanescent field to the external environment [5], [6], [7], [8], [9]. Since Rayleigh signal in the OTDR system is inherently weak, a cleaved D-shaped fiber is developed to conjure the significantly stronger Fresnel signal. While there exist a study that incorporated D-shaped with OTDR as the interrogation system for refractive index sensing [10], temperature sensing using this exact configuration is still yet to be reported. This indicates a clear research gap where D-shaped fiber's evanescent field sensitivity potential to improve temperature response to an OTDR-based sensor remain unexplored. This thesis addresses this gap by developing and experimentally characterizing an OTDR-based temperature sensor employing a D-shaped fiber as the sensing element.

1.3 Aim and Objectives

The main aim of this project is to develop an optical time-domain reflectometer-based temperature sensor employing a D-shaped optical fiber as the sensing element, and to evaluate its performance over a temperature range of 40 ℃ to 80℃.

To achieve this aim, the following objectives are set:

1. To fabricate and characterize D-shaped optical fiber using a side-polishing technique with variants of polishing depths.
2. To investigate the sensing sensitivity, and linearity of D-shaped fiber sensing probe for temperature sensing with OTDR as the main interrogation instrument.

1.4 Scope of the Study

This study focuses on the development and characterization of a point temperature sensor employing a D-shaped single-mode optical fiber (SMF-28) fabricated via side-polishing technique as the sensing element, interrogated by a commercial OTDR operating at 1550nm. The temperature measurement is conducted over a range of 40 ℃ to 80 ℃. Five D-shaped fiber probes with varying polishing depths, characterized by insertion loss levels of approximately 2 dB, 4 dB, 6 dB, 7dB and 8 dB are fabricated and evaluated. The results are then compared with the control bare fiber with no structural modification to evaluate the sensing performance of the D-shaped fibers. This study also explore the feasibility of the sensing element in a static thermal environment by conducting a stability test over a 20-minute period. The study is limited to single-point sensing at a defined position along the fiber, and does not extend to quasi-distributed or fully distributed temperature sensing configurations.

1.5 Significance of the Study

This study presents the first experimental demonstration of a cleaved D-shaped optical fiber sensing element fabricated via side-polishing technique integrated with a commercial OTDR for temperature sensing. By combining the evanescent field

sensitivity of the D-shaped fiber geometry with the practical advantages of OTDR interrogation, this works establishes a new sensing configuration that has not been previously reported. Simultaneously, this study investigates the ideal polishing depth of the D-shape structure that contribute to the highest responsivity to the thermal environment. The findings of this study contribute to the broader field of optical fiber sensing by demonstrating a structurally simple, and practically deployable point temperature sensor architecture.

1.6 Thesis Organization

This thesis is presented in five chapters. This chapter introduce the background of the temperature OFS, outlining the gap in the current literature, together with the scope, significance, aim and objectives of this study. Chapter 2 presents a structured literature reviews, covering the foundations of optical fiber for temperature sensing, theoretical principles and applications of OTDR, waveguide and evanescent field theory, and D-shaped fiber configuration for sensing applications. Chapter 3 describes the methodology used in this studyt, starting with the fabrication process of the bare and D-shaped fiber, the experimental setup of the OTDR-based temperature sensing, and the procedures used for temperature-dependent measurements and data acquisition. Chapter 4 presents and discusses the experimental results, including the OTDR reflection peak response due to the D-shaped fiber acting as a sensing element in response to the temperature of the external environment, analysis of sensitivity and linearity, and a comparative discussion of the sensor's performance relative to existing works. Chapter 5 concludes the thesis by summarizing the principal findings, identifying the limitations of the present work, and recommending directions for future research.

Chapter 2: Literature Review

2.1 Introduction

Optical fiber sensors have emerged as a promising alternative to conventional electronic sensors for a wide range of physical measurement applications. Among the physical parameters that optical fiber sensors are used for measurement, temperature is one of the most significant parameter due to its wide applications across engineering and scientific disciplines.

The sensing mechanisms reported in the literature can be classified based on the property of light that is modulated by the property. Wavelength-modulated sensors, such as FBG sensors, detect temperature-induced shifts in a reflected wavelength. It is one of the most established and widely adopted optical fiber sensing technologies. Phase-modulated sensors, including Mach-Zehnder and Fabry-Pérot interferometers, offer extremely high sensitivity by detecting optical path length changes induced by temperature. Attenuation-modulated, or intensity-based sensors, operate by detecting changes in the optical power transmitted through or reflected from the sensing element, which offer a simpler and more cost-effective alternative to the former categories. Each of these sensing mechanism carries its own trade-offs in terms of sensitivity, interrogation complexity, and cost, which are reviewed in detail in Section 2.2.

Other than sensing mechanism, the choice of interrogation platform also determines how the reliable is the sensor signal. Optical Time-Domain Reflectometer (OTDR), initially developed as a diagnostic tool for telecommunication fiber characterization, has found increasing application as a sensing interrogation platform

due to its practical advantages: single-ended fiber access, commercial availability, and the ability to precisely localize optical loss events at specific positions along the fiber from a single measurement point. OTDR has a promising potential in intensity-based point temperature sensing application particularly when paired with a modified fiber sensing element. The operating principles of the OTDR, the Fresnel reflection-based sensing technique relevant to this work, and a review of prior OTDR-based temperature sensing demonstrations are covered in Section 2.3.

The sensitivity of evanescent field-based OTDR sensor is fundamentally governed by the geometry of the sensing element and the extent to which the guided optical field is exposed to the surrounding environment. In a standard unmodified single-mode fiber, the thick silica cladding completely isolated the guided light from the external environment, which make it insensitive to changes in surrounding medium. Controlled geometric modifications of the fiber, specifically by removing the cladding through side-polishing to produce a D-shaped cross-section, bring the evanescent field of the guided light to within micrometers of the fiber surface, enabling interaction with the surrounding medium or a deposited coating material. Understanding this sensing mechanism requires the fundamental of waveguide theory, total internal reflection, and evanescent field physics, which are discussed in Section 2.4. Furthermore, the fabrication, physics, and sensing applications of D-shaped fiber are reviewed in detail in Section 2.5.

2.2 Temperature Sensor Based on Optical Fiber

2.2.1 Wavelength-Modulated Temperature Sensor

Wavelength-modulated sensors are some of the most established type of OFS, offering extremely high precision for specific, localized applications. These sensors

measure the environment's parameter by detecting the shift in the reflected wavelength. Fiber Bragg Gratings (FBGs) are among the most common wavelength-modulated sensors and are widely used for temperature and strain measurement. The sensing principle relies on a periodic perturbation of the refractive index, hence the name 'Gratings', along the fiber core. These gratings typically fabricated by exposing the core of the fiber to an interference pattern of UV light. This refractive index perturbation essentially functions as a narrowband reflection filter. When broad-spectrum light is transmitted through the fiber, the majority of wavelengths pass through, while only a specific wavelength is reflected known as Bragg wavelength [11]. At the Bragg wavelength, reflections from the periodic perturbations interfere constructively to create a strong reflection signal. This relationship is governed by the Bragg condition:

$$\lambda_B = 2\, n_{eff}\, \Lambda \tag{1}$$

where λ is the Bragg wavelength, n_{eff} is the effective refractive index, and Λ represents the grating period. Environmental factors directly affect the value of n_{eff} and Λ, which result in wavelength shift, useful for quantifying the physical parameters [11]. For temperature sensing, two physical mechanisms contribute to this wavelength shift: the thermo-optic effect, which changes n_{eff} with temperature, and thermal expansion, which changes the grating period Λ [12]. In a standard single-mode fiber, combinations of these mechanisms yield a temperature sensitivity of about6.8-13 pm/℃ [11]. Geometric modifications, such as tapering, has been shown to not affect the thermal sensitivity of this type of OFS [13]. However, it is possible to increase the sensitivity beyond the intrinsic silica response by using specialized fiber. For instance, lead sulfide (PbS)-doped silica FBGs have shown to possess sensitivities

of approximately 19 pm/ ℃ over a range of 20-150 ℃, which is an enhancement of about 60% over standard SMF FBG sensor [14]. However, such approach is more complex and costly.

One of the most cited limitation of FBG temperature sensor is the cross-sensitivity between temperature and strain. This phenomenon happened due to both temperature and strain contribute to the shift in the Bragg wavelength, making it difficult to identify the perturbation source. To overcome this, an additional reference gratings often have to be incorporated into the sensing system to isolate the strain and temperature data. The system also required an expensive interrogation system capable of detecting shift within sub-picometer range with sufficient resolution [15]. Despite these limitations, FBG compatibility with wavelength-division multiplexing (WDM) makes it particularly capable of multi-point sensing, allowing for a quasi-distributed measurement over the entire length of the fiber. Zhang and colleagues [16] have demonstrated a 21-channel FBG array capable of real-time temperature field monitoring with a spatial resolution of 3 mm and a temperature sensitivity of 9.7 pm/℃. Zhong and colleagues[17] also have developed an FBG sensor with 18 temperature sensing points and another 60 strain sensing points capable of spatial resolution less than 1 mm.

2.2.2 Phase-Modulated Temperature Sensor

Phase-modulated sensors offer a different approach to optical fiber sensing, where the measured optical parameter is the phase shift or interference pattern of the light propagating through the fiber, rather than the shift in the Bragg wavelength. There are multiple configurations for this type of OFS, including Mach-Zehnder, Sagnac, Michelson interferometer but the most widely research for temperature

sensing is the Fabry-Perot interferometer (FPI). A fiber-optic FPI is formed by creating two partially reflective surfaces separated by a cavity length L, which may be an air gap, a polymer-filled region, or a section of specialty fiber. Light that enter the cavity undergoes multiple reflections between the two surfaces, producing an interference spectrum with fringe spacing and position that are both sensitive to changes in the optical path length nL, where n is the refractive index of the cavity medium [18]. Temperature changes alter the cavity length through thermal expansion and the cavity refractive index through the thermo-optic effect, shifting the fringe pattern in a measurable and predictable manner.

The main advantage of FPI sensors for temperature measurement is their ability to achieve very high sensitivities by choosing cavity materials with large thermo-optic coefficient. Simple air-cavity FPIs formed from standard single-mode fiber exhibit temperature sensitivities comparable to FBGs, whereas polymer-filled cavities, such as polydimethysiloxane (PDMS), exhibit much higher responsivity [19]. A miniature fiber-optic FPI sensor employing a UV-glue-filled cavity demonstrated a temperature sensitivity of 1.174 nm/ ℃ with a highly linear response in the range of 30-60℃ [20]. Further enhancement is achievable through the Vernier effect, where two FPIs with slightly mismatched free spectral range are cascaded or placed in parallel to produce an amplified envelope response. Using the Vernier effect, a temperature sensitivity of -3.38 nm/ ℃ has been demonstrated, approximately 8.45 times higher than a single FPI in the same configuration [21]. These figures represent sensitivities several orders of magnitude higher than standard FBGs, making FPI sensors attractive for high-resolution, localized temperature measurements.

However, these sensitivities advantage necessitates an advance spectral interrogation system that able to resolve minuscule fringe shifts such as an optical spectrum analyzer and tunable laser. This makes FPI particularly incompatible with the low-cost, intensity-based interrogation approach central to this work. Furthermore, FPI is fundamentally limited to single-point sensing. Scaling it to a quasi-distributed measurement via WDM will increase the complexity of the system due to parasitic interference between the sensors. This situation is inherently due to the fact that only a single wavelength of light is used in the FPI sensors. A signal from one sensor will introduce noise to other sensors and vice versa by the power of two. It also has been shown that multiplexing the sensor will degrade the resolution by five times compare to the single-point sensor for this configuration [22].

2.2.3 Attenuation-Modulated Temperature Sensor

Attenuation-modulated optical fiber sensors operate on a more straightforward principle than the former two types of sensors, where the magnitude of optical loss is measured and correlated to the changes in the environmental parameters. In the context of temperature sensing, the inherent temperature sensitivity of standard silica fiber attenuation is typically too low for practical measurements. Therefore, to create a highly responsive attenuation-modulated sensor, the fiber must be structurally modified and paired with a temperature-sensitive material. By altering the fiber geometry to expose the guided light, the evanescent wave can interact directly with a thermo-optic coating applied externally.

OTDR is an ideal interrogation technique for attenuation-modulated sensors. It operates by launching a narrow optical pulse into the fiber and measuring the intensity of the Rayleigh backscattered light over time. Since the time of flight correlates

directly to distance, OTDR provides a spatial map of attenuation along the entire length of the fiber [23]. When configuring a remote point sensor, the D-shaped sensing zone is placed at a specific target location at the end of the fiber. As the ambient temperature at the specific location fluctuates, the refractive index of the immediate external environment changes. This induces a localized variation in optical loss that the OTDR detects as a discrete attenuation peak due to Fresnel reflection on the trace as illustrated in **Fig. 2.1 [24]**. By monitoring the magnitude of this specific peak, the OTDR can precisely track the temperature at the remote point.

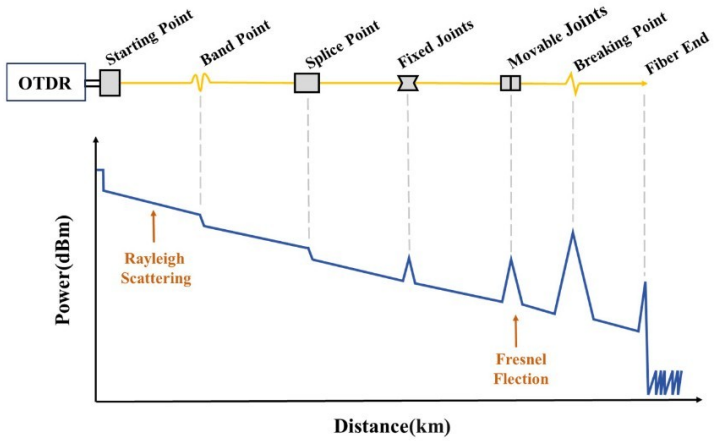


Fig. 2.1: Typical OTDR test curve showing Rayleigh scattering and Fresnel reflection Peaks [19].

The sensitivity of an attenuation-modulated sensor heavily depends on the thermo-optic coefficient of the material interacting with the evanescent field. Materials with high thermo-optic coefficients experience significant changes in their refractive index or absorption properties as temperature fluctuates. Polymers such as PDMS are frequently utilized due to their large negative thermo-optic and thermal expansion coefficients. When temperature rises, the refractive index of the PDMS coating decreases, which alters the boundary conditions at the sensing interface and modulates the optical loss of the guided light[25]. Beyond polymers, novel nanomaterials have shown a potential for attenuation-based temperature sensing.

Mohanraj and colleagues [8] demonstrated that coating the flat surface of a D-shaped fiber with few-layer Molybdenum disulfide (MoS_2) nanosheets yields an ultrasensitive temperature sensor. Listewnik and colleagues [26] have reported applying ZnO ALD coating on a microsphere-based optical fiber temperature sensor. The result indicate that sensor with a higher coating thickness of 200 nm possessed a superior sensitivity than the 100 nm counterpart, with sensitivity of each fiber to be 103.5 nW/ ℃ and 11.4 nW/℃ respectively.

When developing an OTDR-based point sensor, the primary design focus is optimizing the signal-to-noise ratio (SNR) and detection accuracy over the required transmission distance. The geometry of the D-shaped fiber, specifically the polishing depth and length must be engineered so that the temperature-dependent attenuation is prominent enough to exceed the OTDR's intensity resolution limit. Furthermore, the parameters of the OTDR itself play a critical role in sensor performance. A shorter pulse provides a higher spatial resolution, allowing the system to accurately pinpoint the exact location of the sensor, but it carries less optical energy. Reduced energy decreases the dynamic range and SNR, which can limit the maximum distance the point sensor can be placed from the interrogator. Conversely, a longer pulse carries more energy for long-distance remote sensing but come with a lower spatial resolution. Therefore, the interrogation pulse width (PW) must be chosen carefully to achieve high-resolution, long range temperature monitoring [23].

2.3 Principles and Applications of Optical Time-Domain Reflectometer (OTDR)

2.3.1 OTDR Operating Principles

OTDR was first reported by Barnoski and Jensen [27] in 1976, who demonstrated that the time-dependent profile of light backscattered within an optical fiber waveguide could be recorded and used to determine the fiber's distributed attenuation characteristics from a single access point at the fiber input. Five decades since that foundational work, OTDR has become the dominant instrument for characterizing, and troubleshooting optical fiber networks worldwide. Its transition from a pure telecommunications diagnostic tool into a platform for distributed physical sensing is a more recent development. Knowledge of OTDR operating principle is essential to understand both its capabilities and its limitations as a sensing interrogation system.

OTDR in SMFs operate based on the mechanism of Rayleigh scattering. During the manufacturing process of silica-based optical fibers, microscopic density fluctuations are permanently frozen into the glass matrix. As the optical pulse propagates through the core, these small variations caused a fraction of the optical power to scatter in all directions. A very small portion of this scattered light falls within the numerical aperture of the fiber and is guided back toward the source as backscattered light [1].

The power of the backscattered signal $P_S(L)$ received by the OTDR detector for a specific distance L along the fiber can be mathematically modeled by the standard OTDR equation:

$$P_S(L) = S \cdot \alpha_s \cdot W \cdot P_0 \cdot e^{-2\alpha L} \tag{2}$$

where S is the backscatter capture fraction, which is the proportion of scattered light successfully guided back through the core, a_S is the Rayleigh scattering coefficient, a is the total intrinsic attenuation coefficient of the fiber, and P_0 is the initial power of the injected optical pulse. Crucially, W represents the spatial length of the propagating pulse, which is directly proportional to the duration of the laser pulse τ. It is defined as:

$$W = \tau \cdot v_{gr} = \tau \frac{c}{n_{gr}}, \tag{3}$$

where τ is the temporal pulse duration, v_{gr} is the group velocity of the light, c is the speed of light in a vacuum, and n_{gr} is the group index of the glass [28].

An OTDR trace plots this backscattered power $P_s(L)$ on a logarithmic scale against the distance L. In a uniform, unperturbed fiber, this trace appears as a linear decay, where the constant downward slope visually represents the fiber's baselines intrinsic attenuation. Localized physical perturbations that induce sudden optical loss will appear as a distinct vertical drops or peaks on the trace, providing the operational basis for attenuation-modulated sensing.

Two critical performance parameters govern OTDR performance: dynamic range and spatial resolution. The dynamic range and spatial resolution. The dynamic range defines the maximum optical loss the OTDR can overcome, effectively determining the maximum measurable fiber length. It is quantified as the difference between the initial backscatter level and noise floor. The spatial resolution, on the other hand, determines the minimum distance between the two events that can be distinguished as separate features. It can be mathematically represented with the equation:

$$\Delta z = \frac{v_g \tau}{2}. \quad (4)$$

A shorter PW yields finer spatial resolution but at the cost of reduced backscattered energy, hence a shorter dynamic range. Conversely, a longer pulse injects more energy into the fiber, essentially extending the measurement range but degrading the ability to resolve closely spaced events [24]. This inherent trade-off between dynamic range and spatial resolution represents a fundamental design constrain in conventional OTDR system.

In the context of sensing applications, Rayleigh-based OTDR has been widely used for distributed temperature monitoring by detecting changes in the backscattered signal intensity along the fiber. However, the Rayleigh backscatter signal form a standard single-mode fiber is relatively weak, ranging from -40dB to -80dB relative to the launched pulse power which limits the signal-to-noise ratio for high-precision sensing [29]. This limitation motivates the exploration of alternative sensing approaches, such as Fresnel reflection-based techniques, which leverage the significantly stronger reflections at discrete fiber interfaces to achieve enhanced sensitivity, as discussed in the following section.

2.3.2 Fresnel Reflection-Based Sensing Technique

While Rayleigh scattering provides a continuous, distributed signal along the fiber length, an alternative sensing approach leverages discrete, high-intensity reflections that occur at well-defined interfaces within the optical path. This technique, known as Fresnel reflection-based sensing, offers significantly higher signal-to-noise ratios and simplified signal processing, making it particularly attractive for point sensing applications such as the one proposed in this thesis.

Fresnel reflection is a fundamental optical phenomenon that occurs when light propagating through a medium encounters an abrupt change in refractive index. At such interface, a portion of the incident optical power is reflected toward the source, while the remainder is transmitted across the boundary. For the case of a cleaved optical fiber endface, where the fiber core (with refractive index n_f) meets an external medium (with refractive index n_m) at normal incidence, the fraction of reflected power, known as reflectance R, is given by the Fresnel equation [30], [31]:

$$R = \left(\frac{n_f - n_m}{n_f + n_m}\right)^2. \quad (5)$$

For a standard silica single-mode fiber ($n_f \approx 1.4682$ for λ=1550nm) terminated in air ($n_m \approx 1.000$), the reflectance at normal incidence is approximately 3.6% or about -14 dB in return loss. By contrast, Rayleigh backscattering in SMF-28 is extremely weak, with Rayleigh backscattering of -77 dB at 1310 nm and -82 dB at 1550 nm [32]. This indicates a glass-to-air Fresnel reflection can be thousands of times stronger than the Rayleigh backscatter signal. This orders-of-magnitude difference in signal strength is the primary advantage of Fresnel-based sensing over Rayleigh-based approaches.

The sensing mechanism is realized by detecting change in the refractive index of the external medium, which directly affecting the Fresnel reflection R, hence the amplitude of the reflection peak measured by the OTDR. For temperature sensing, this principle is influenced by the thermo-optic effect exhibited by the external medium. Specifically, when the fiber end is placed in contact with medium with high thermo-optic coefficient, the variance of n_m can be exploited for temperature measurement. This relationship is expressed as:

$$T = \left(\frac{\partial n}{\partial T}\right)^{-1} n_m + C_0, \tag{6}$$

where $\partial n_m / \partial T$ is the thermo-optic coefficient of the medium, n_m is the refractive index of the medium, and C_0 is the offset constant for the reference refractive index. Materials such as epoxy resin, which exhibits a large thermo-optic coefficient and a refractive index closely matched to that of silica, have been successfully employed as Fresnel sensing media at fiber endfaces [33].

A key advantage of the Fresnel reflection approach in OTDR-based systems is that the position of the sensing point can be precisely identified from the time-domain position of the reflection peak on the OTDR trace. This enables both the measurement of the parameter of interest and the unambiguous localization of the measurement point using a single, commercially available instrument. Yuan and colleagues [34] have demonstrated a remote refractive index sensor that combined Fresnel reflection with OTDR interrogation, achieving measurement distances of up to 100.8 km with sensitivities ranging from 38.71 dB/RIU to 304.89 dB/RIU over a refractive index of 1.3486 to 1.4525. In their configuration, the Fresnel reflection from a cleaved single-mode fiber endface immersed in a test sample was read directly from the OTDR trace, with the peak height serving as the measurand.

Several methods have been developed to improve the practicality and robustness of Fresnel reflection-based sensors. To eliminate errors caused by fluctuations in the light source power or variations in fiber attenuation with distance, a reflective intensity technique is commonly employed, where the Fresnel reflection signal is normalized against a reference signal. In the works of Chen and Huang [33], a two-channel configuration using three fiber couplers and two photodetectors allowed

the reference and sensing signals to be compared, yielding a temperature measurement with an R^2 linearity of 0.9982 across a temperature range of 50 to 120 ℃. They and other colleagues [35] then further extended this approach to a parallel-multipoint configuration using an arrayed waveguide grating (AWG) for wavelength-division demultiplexing, achieving a temperature resolution of approximately 0.03 ℃.

2.3.3 Previous Works on OTDR-Based Temperature Sensing

OTDR systems are highly valued due to its capability of long-range, remote sensing over a single strand of fiber, utilizing either intrinsic scattering mechanisms or structurally induced localized attenuation events. The most commercially established distributed temperature sensors (DTS) rely on Raman and Brillouin scattering. Raman OTDR (ROTDR) isolates the temperature-dependent anti-Stokes scattering and compares it against temperature-sensitive Stokes scattering to yield a temperature-insensitive Stokes scattering to yield a temperature profile. Advanced ROTDR systems have demonstrated the capacity to measure temperatures with a ℃5 resolution over ranges spanning 15 to 25 kilometers. Brillouin OTDR (BOTDR) and Brillouin optical time-domain analysis (BOTDA) are also widely use to track the temperature and strain-induced shifts in the Brillouin frequency [36]. While exceptionally effective for long-distance distributed structural monitoring, these scattering techniques often suffer from low SNR, which requires long acquisition times, complex signal processing, and expensive optical components like narrow-linewidth lasers or high-speed demodulators [37]. Furthermore, researchers have recently developed hybrid OTDR systems that multiplex Raman scattering and Rayleigh scattering to achieve dual-parameter monitoring of temperature and vibration [38].

Conventional OTDR provides higher spatial resolution than Raman or Brillouin systems but is inherently limited by the very weak temperature dependence of standard silica glass. To overcome this and enhance the Rayleigh backscatter amplitude for temperature sensing, early works modified the core materials of the sensing fiber. Hartog [39] has replaced the solid core with a higher refractive-index liquid, which exhibited a much stronger thermal response with accuracy of ±1 ℃ and a spatial resolution of 1 m over 100 m fiber length. However, the liquid core limited its practical application in extreme environments. Farries and colleagues [40], [41] also have reported OFS utilizing fibers doped with rare-earth elements. By tracking attenuation changes near the edge of these bands via OTDR, they achieved a 2 ℃ accuracy with 15 m spatial resolution over 200 m using Neodynium-doped (Nd^{3+}) fibers, and improved this to 1℃ accuracy with 3.5 m resolution using Holmium-doped (Ho^{3+}) fibers. Jin and colleagues [37] have developed a low-cost distributed temperature sensor using Germanium-Phosphorus (Ge/P) co-doped fibers subjected to gamma radiation. The radiation-induced attenuation demonstrated stable temperature dependence after sufficient annealing, achieving a temperature resolution of 0.05 ℃ and a spatial resolution of 16.3 m over a 100 m length. Furthermore, Loranger and colleagues [42] demonstrated that simple localized UV exposure on standard single-mode fibers could permanently increase the Rayleigh scattering by an order of magnitude, which improve the SNR and resolution of distributed temperature measurements.

To achieve ultra-high sensitivity, researchers transitioned from simple-intensity based Rayleigh OTDR to coherent phase-sensitive OTDR (Φ-OTDR) and Optical Frequency-Domain Reflectometry (OFDR).These systems detect the relative phase

changes between multiple Rayleigh scattering centers induced by temperature variations. By employing highly coherent lasers and cross-correlation algorithms, methods such as chirped-pulse Φ-OTDR can directly measure distributed temperature-induced time delays, realizing exceptional temperature accuracies as high as 0.028 ℃ [29].

While distributed sensing monitors the entire fiber length, OTDR is also highly effective for interrogating discrete, localized point sensors based on engineered attenuation or multimode interference. One such example is the work by Hatta and colleagues [43] who developed an OTDR-interrogated temperature sensor based on a Singlemode-Multimode-Singlemode (SMS) fiber structure. In their design, a segment of multimode fiber acts as the sensing zone as illustrated in **Fig. 2.2**. Ambient temperature changes alter the refractive index and dimensions of the multimode core and cladding, which modulates the multimode interference and subsequently changes the return loss of the transmitted light. The OTDR identifies this localized structural change as a distinct, non-reflective attenuation event. By tracking the magnitude of this return loss, they achieved a temperature measurement range of 40 ℃ to 195 ℃ with a resolution of 0.1℃ using a 60 mm sensing length. Furthermore, they demonstrated that multiple SMS structures could be cascaded along the fiber, proving that conventional OTDR instruments can robustly interrogate discrete optical power-loss sensors for remote point-sensing applications.

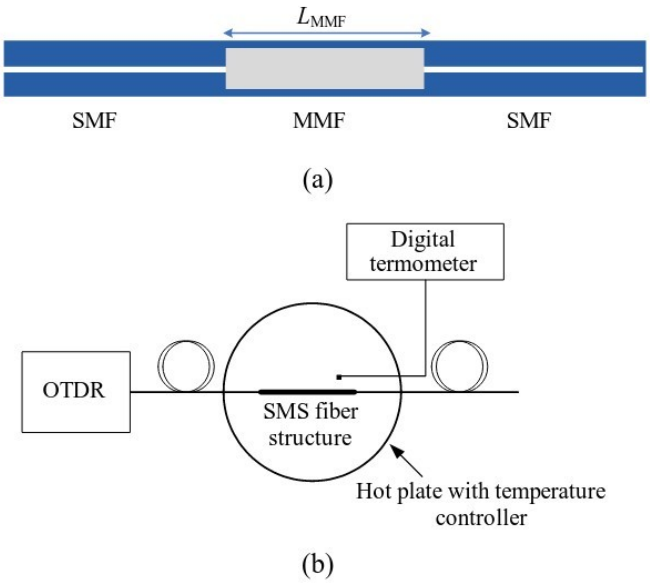


Fig. 2.2: OTDR-based sensor for temperature monitoring: (a) SMS (Singlemode-Multimode-Singlemode) fiber structure (b) schematic setup for measuring the temperature variation using an OTDR [43].

2.4 Physics of Optical Fiber

2.4.1 Propagation of Light in Optical Fiber

The propagation of light within an optical fiber is governed by the Total Internal Reflection (TIR), which is determined by how light act according to the Snells' law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \tag{7}$$

where θ_1 is the angle of incidence, θ_2 is the angle of refraction, n_1 is the refractive index of the core, n_2 is the refractive index of cladding. For TIR to occur, list must be travelling from a medium of higher refractive index (core) to one of lower refractive index (cladding). Within the fiber, the incident angle at the core-cladding interface must be equal to or greater than the critical angle θ_c which is defined as the angle of incidence for which the angle of refraction equals 90° [30]. Critical angle is given by:

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right). \tag{8}$$

When the angle of incidence is greater than θ_c, $\sin\theta_2 > 1$, which has no real solution. In such case, the incident light is perfectly reflected into the core, enabling lossless guidance along the fiber. **Fig. 2.3** illustrated TIR in different mediums.

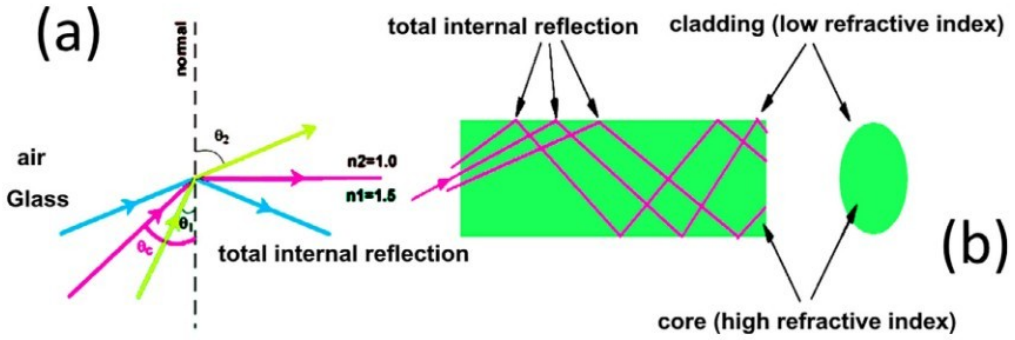


Fig. 2.3: Total Internal Reflection (TIR) illustrated in (a) glass and air mediums and (b) a fiber structure with different refractive indices [44].

The ability of an optical fiber to capture and guide light is quantified by its numerical aperture (NA). Consider a ray incident on the fiber end-face from air with refractive index $n_0 \approx 1$. For the fiber to act as a waveguide, the light need to refract into the core at an angle that satisfied the TIR condition at the core-cladding interface. The maximum acceptance angle θ_a, which is the largest angle at which light can enter the fiber and still be guided, is derived from Snell's law applied at both the air-core interface and the core-cladding interface, resulting in [30]:

$$NA = n_0 \sin\theta_a = \sqrt{(n_1^2 - n_2^2)}\,. \tag{9}$$

For standard single-mode fiber, with a core index of approximately 1.468 and a cladding index of approximately 1.462 at 1550 nm, the NA is approximately 0.14, corresponding to an acceptance angle 8° in air. The guided power fraction is strongly dependent on the launch angle, with optimal confinement observed at low angles [45]. A higher NA indicates a larger acceptance cone and therefore more efficient light

capture, but also implies a larger index contrast, hence a greater number of supported modes.

However, TIR alone is not sufficient to guarantee single-mode operation. Single-mode guidance in an optical fiber is only possible when the normalized frequency, also called the V-number, is below a critical threshold. V-number is given by:

$$v = \frac{2\pi a}{\lambda_0}\sqrt{(n_1^2 - n_2^2)}, \tag{10}$$

where a is the fiber core diameter, λ_0 is the light propagation wavelength. For step-index fibers, V must be less than 2.405 to operate in single-mode. If the normalized frequency exceeds this value, higher-order modes begin to propagate and the fiber will operate as a multi-mode fiber (MMF). As a result, the pulse that reach the end of the fiber will have a slight distortion [44].

2.4.2 Thermo-Optic Effects in Silica Fiber

The thermo-optic effect is the physical mechanism that enables the conversion of a thermal stimulus into an optical signal in the OTDR-based temperature sensor. When the temperature of the fiber changes, the refractive indices of the core and cladding change accordingly. As a result, it alters the propagation characteristics of the guided modes, hence the backscattered signal detected by the OTDR. The thermo-optic coefficient, $\partial n/\partial T$, describes the rate of change of refractive index with temperature. For silica glass, this coefficient has two contributing mechanisms: thermal expansion and electronic polarizability [46].

Thermal expansion describes how the glass matrix physically expands as temperature increases, reducing the atomic density of the material and consequently lowering the refractive index. Electronic polarizability, on the other hand, describes how changes in the temperature alter the vibrational state of the atoms in the glass network. These atomic vibrations directly modify the electronic structure of the glass, changing how strongly the material polarizes in response to and applied electromagnetic field, hence modifying the refractive index [46]. For pure $_2$SiO thermal expansion accounts for only approximately 2% of the total coefficients, with the remaining ~98% contributed by the temperature-dependent change in electronic polarizability [46]. This confirms that the thermo-optic effect in silica is primarily an electronic phenomenon, not a structural one.

Gao and colleagues [47] have investigated the thermo-optic coefficient of silica fiber over a wide range of temperature using an extrinsic Fabry-Perot interferometric method. They found the first-order coefficient to be 1.090×10^{-5}/℃ and second-order to be 1.611×10^{-9}/℃^{2}. While linear model of temperature dependence of the refractive index provide a good approximation at lower temperature, the quadratic model is more rigorous over wide temperature ranges. Despite having a smaller value, the second-order coefficient is not negligible and becomes increasingly significant at elevated temperatures. The effective refractive index of the fiber therefore can be expressed as [47]:

$$n(T)=n_0+\alpha_n T+\beta_n T^2, \tag{11}$$

with n_0 is the refractive index at a reference temperature, T is the temperature, a_n is the first-order thermo-optic coefficient and β_n is the second-order thermo-optic coefficient.

For comparison, water has a thermo-optic coefficient of -1.12×10 $^{-4}$/℃ at 1550 nm [48], which is negative and approximately one order of magnitude larger than that of silica. This stark contrast arises because the refractive index of water, unlike silica, decreases with increasing temperature.

It is also important to note that the thermo-optic coefficient of the fiber core may differ slightly from that of the pure silica cladding due to the presence of dopants. In the case of Corning SMF-28, the core is doped with GeO $_2$ to raise the refractive index above that of pure silica cladding. Rego [49] estimated the thermo-optic coefficient of bulk GeO $_2$ to be approximately 18.3×10 $^{-6}$ K^{-1} at 293 K and 1550 nm, which is notably higher than pure SiO_2.

2.5 Fabrication and Sensing Principle of D-Shaped Fiber

2.5.1 Fabrication and Geometry of D-Shaped Fiber

A D-shaped optical fiber is defined by an asymmetric cross-sectional profile in which one side of the outer cladding has been removed or formed flat, leaving the remaining structure with a geometry resembling the letter "D" as shown in **Fig. 2.4**. This flat surfaces allow the evanescent field of the guided light to interact directly with the surrounding external medium. The fabrication of D-shaped fibers requires a high precision technique to ensure that the core is not damaged while enough cladding is removed to expose the evanescent field. In particular, there are three fabrication methods often mention in the literature: mechanical side-polishing, chemical etching, and femtosecond laser machining.

Mechanical side-polishing is the most widely adopted method for creating D-shaped fibers due to its cost-effectiveness, environmental safety, and high degree of

control. In this method, a section of stripped single-mode fiber is securely held in place by high-precision optical alignment stages while a motor-driven cylindrical shaft or grinding wheel applies friction to one side of the fiber. The grinding wheel is typically covered with abrasive material, such as sandpaper or diamond bit [10], [50]. To achieve a high-quality optical finish, researchers often employ a multi-step polishing process. For example, the fiber may first be roughly ground down using aluminum oxide (Al_2O_3) powder, followed by ultrasonic cleaning in deionized water, and finally polished with cerium oxide (CeO_2) powder to minimize surface roughness [7].

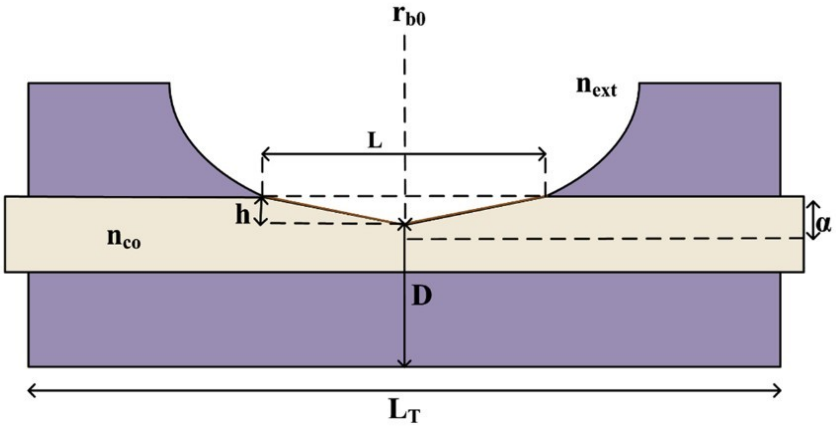


Fig. 2.4: Schematic diagram of a D-shaped fiber showing its asymmetric geometry [2].

D-shaped fiber can also be fabricated by means of chemical etching. For this method, hydrofluoric acid (HF) aqueous solution is used to locally thin the fiber structure by removing the cladding layer. Quero and colleauges [51] have demonstrated this fabrication method by locally etched a fiber with a 6% HF aqueous solution at room temperature and subsequently neutralized with CaO as the basic solution.

For applications requiring micro-scale geometric precision, femtosecond laser systems are utilized. This approach uses highly focus, ultra-short pulses to remove the

silica material layer by laser. Zhu and colleagues [5] demonstrated D-shaped fabrication by using a femtosecond laser operating at 515 nm wavelength with a 290 fs pulse duration to perform cyclic etching on a single-mode fiber. The laser processing allowed for the creation of a highly controlled D-shaped groove with reduction in surface roughness. For this study, the mechanical side-polishing method was used due to it being very cost effective while less hazardous compare to the chemical etching method.

The sensitivity of a D-shaped fiber sensor is directly related to its geometry, specifically the residual cladding thickness, which is the shortest distance between the flat surface and the outer boundary of the fiber core. Decreasing the residual cladding thickness exponentially increases the evanescent wave into the external medium. For instance, Kim and colleagues [7] demonstrated that D-shaped fibers with remaining cladding thickness of 2.8 μm, 1.5 μm, and 0.3 μm exhibited transmission losses of 0.5 dB, 1.0 dB, and 1.5 dB, respectively. The geometric design of the D-shaped sensing zones must strike a critical balance where the residual cladding thickness must be thin enough to allow for sufficient evanescent interaction with the external medium while maintaining the structural integrity of the optical fiber itself.

2.5.2 Evanescent Field and Modified Lambert-Beer Law

While TIR described light reflection in the core waveguide to be fully reflected back to the core, electromagnetic theory offer a more nuanced description crucial for sensing purpose. Maxwell's equation determine that the electromagnetic field cannot discontinue abruptly at the core-cladding interface. Instead, a portion of the field penetrates into the lower refractive index medium (the cladding), also known as the evanescent field. Evanescent field propagates parallel to the core-cladding interface

and its amplitude decays exponentially with distance x from the core boundary, given by:

$$E(x)=E_0\exp(\frac{-x}{d_p}), \tag{12}$$

where x is the distance from the fiber core, with $x=0$ at the core-cladding interface, E_0 is the magnitude of the field at the interface, and d_p is the penetration depth.. The penetration depth is the distance at which the electric field amplitude decays relative to the interface and is given by:

$$d_p=\frac{\lambda}{2\pi\sqrt{(n_{co}^2\sin^2\theta-n_{cl}^2)}}, \tag{13}$$

where λ is the wavelength of the light source, n_{co} and n_{cl} are refractive index of the core and cladding, respectively, and θ is the angle of the incidence at the core-cladding interface [52]. Several important observations can be made from the expression. First, the penetration depth is directly proportional to the wavelength, which indicates longer wavelengths produce a deeper evanescent field hence why near-infrared wavelengths around 1550 nm are commonly used in evanescent-field sensing applications. Second, the penetration depth increases as the angle of incidence approaches the critical angle ($\sin\theta\rightarrow n_2/n_1$), diverging to infinity at exactly the critical angle, where the field is no longer evanescent but is instead a propagating field in the cladding. Third, a smaller index difference between the core and cladding yields a larger penetration depth.

For single-mode fiber operating at 1550 nm, the penetration depth is in approximately 400 nm, which is substantially smaller than diameter of a standard fiber cladding. The evanescent field decays to absolute zero long before it reaches the outer

surface of the fiber. Therefore, while the evanescent wave provides a theoretical mechanism for light to interact with the environment, accessing this field requires alteration of the physical geometry of the waveguide to bring the external environment within the sub-microscopic range of the penetration depth.

The polished surface of the D-shaped fiber functions as a window for the exponentially decaying evanescent field to penetrate the external medium. However, quantifying the optical attenuation induced by this localized environment requires a modified mathematical framework, as the physics of standard optical absorption do not apply. In standard free-space optics, light attenuation is strictly governed by the standard Lambert-Beer law. However, in a fiber optic evanescent wave (FOEW) sensor, only a fraction of the optical beam pass through the external medium. Therefore, the absorbance (A) is governed by the modified Lambert-Beer's law:

$$A = \alpha L \eta, \tag{14}$$

where a is the absorption coefficient, L is the length of interaction, and η is the fraction power of evanescent field. The absorption coefficient a is fundamentally defined as $\alpha = \epsilon C$, where ϵ is the molar absorptivity and C is the concentration of the target species. By substituting this into the modified law, the sensitivity S for weakly absorbing species is approximately given by:

$$S \approx \epsilon L \eta. \tag{15}$$

This approximation establishes that for a given interaction length L and target medium ϵ, the sensitivity of the sensor is strictly proportional to η *[52]*.

The modified Lambert-Beer law provides the exact theoretical justification for structurally altering the optical fiber. In a standard uniform optical fiber, the fraction of light in the evanescent η is extremely low. For a standard 200 μm diameter fiber, η

is typically in the 10^{-4} range, which result in negligible sensitivity [52]. By polishing the cladding of the fiber, a larger percentage of the electromagnetic is absorbed into the external medium. This directly amplifies η, ensuring that any localized attenuation induced by the environment, such as microscopic changes in refractive index driven by thermal stress, results in a proportionally larger optical loss capable of resolved by an OTDR.

2.5.3 Previous Works on Temperature Sensor Based on D-Shaped Fiber

The application of D-shaped fiber in temperature sensing has garnered significant research focus in the past decade. Since standard silica possesses a very low thermo-optic coefficient, bare D-shaped fibers are inherently insensitive to thermal changes. Consequently, recent works have focused on integrating the D-shaped surface with highly thermo-sensitive coatings, utilizing various optical mechanisms to detect temperature. Many high-sensitivity temperature sensors paired the D-shaped profile with interferometry or fiber gratings. For instance, Zhu and colleagues [5] have coated a D-shaped Mach-Zehnder interferometer (MZI) with PDMS which when heated, the refractive index change shifts the MZI interference spectrum, achieving a temperature sensitivity of up to 5.15 nm/ ℃. Similarly, Kim and colleagues [7] have deposited a surface photoresist long-period grating (PR-LPG) onto a D-shaped photonic crystal fiber (PCF) allowing them to track temperature-induced resonant wavelength shifts from 30 ℃ to 100 ℃. Luo and colleagues [6] have utilized optical fiber Sagnac interferometers containing a D-shaped microfiber immersed in sucrose solutions, taking advantage of the liquid's thermal properties to monitor temperature changes via spectral blue-shifts. D-shaped fiber is also frequently

combined with Surface Plasmon Resonance (SPR) and specialized liquids. Mollah and colleagues [9] have incorporated a thin gold film and benzene as the temperature-sensitive overlay onto the D-shaped PCF. Since benzene's refractive index is highly responsive to heat and closely matches that of fused silica, this plasmonic sensor achieved an exceptional maximum sensitivity of 110 nm/°C. Other comparative studies have demonstrated that filling the channels of a D-shaped fiber with liquid analytes like chloroform or chloroform-ethanol mixtures can provide highly tunable, high-resolution temperature tracking suitable for both biomedical and industrial environments [3]. Of particular relevance to intensity-based monitoring systems are D-shaped fibers that rely on direct optical power loss rather than wavelength tracking. One such example is the development of an all-fiber temperature sensor utilizing a D-shaped fiber coated with few-layer MoS_2 nanosheets by Mohanraj and colleagues [8]. MoS_2 exhibits high thermal conductivity and strong optical absorption. As the ambient temperature increases, the strong interaction between the fiber's evanescent field and the MoS_2 coating modulates the transmitted optical power. This attenuation-based sensor achieved an ultra-high optical power-sensitivity of 0.1211 dB/°C, alongside excellent measurement precision linearity in the 26 to 83°C range. **Table 2.1** compare the temperature sensor based on the D-shaped fiber that has been reported in the literature.

Table 2.1: Comparison of D-shaped fiber-based temperature sensors reported in the literature.

Interrogation Method & Core Fiber Type	Temperature Sensing Range	Sensitivity	Linear Correlation Coefficient	References
Intensity-based transmission power variation using a D-shaped single-mode fiber (HI-1060).	26°C – 83°C	0.1211 dB/°C	R^2=0.9963	[8]
Wavelength shift via an MZI cascaded with a Tilted FBG on SMF (G6520D).	20°C – 60°C	4.38 to 5.15 nm/°C	R^2=0.9932 – 0.9998	[5]
Wavelength shift via a surface PR-LPG on a side-polished PCF.	30°C – 100°C	-0.3 nm/°C	-	[7]
Wavelength shift via high-birefringence D-shaped microfiber embedded within a Sagnac Loop Interferometer.	22°C – 60°C	-0.98921 nm/°C	-	[6]
Wavelength interrogation tracking SPR peak shift on a D-shaped PCF	10°C – 70°C	Up to 110 nm/°C	R^2=0.9688 – 0.9814	[9]

Chapter 3: Methodology and Experimental Setup

3.1 Experimental Overview and System Architecture

This chapter describes the design and implementation of the experimental setup used to characterize the optical loss response of D-shaped fiber sensing probes as a function of temperature. The methodology flow of the this study is as illustrated in **Fig. 3.1**.The setup is designed with an OTDR as the main interrogation tools with 1550nm optical pulse launched into the measurement chain and the backscattered return signal is analyzed as a function of round-trip time to produce a spatially resolved loss profile along the fiber length. The D-shaped single-mode fiber probe is placed at a predetermined position where the temperature-dependent evanescent loss appears as a measurable reflective peak on the OTDR trace. The magnitude of the peak at the probe location is the main variable measured throughout this experiment.

The complete measurement setup is as illustrated in **Fig. 3.2**. It consists of three functional segments connected in series: the OTDR instrument, a 2 km single-mode fiber launch spool, and the D-shaped sensing probe terminated at its far end by an open-cleaved fiber face. Each segment serves a distinctive role throughout the measurement process to satisfy the specific requirements for OTDR interrogation and evanescent sensing, as detailed below.

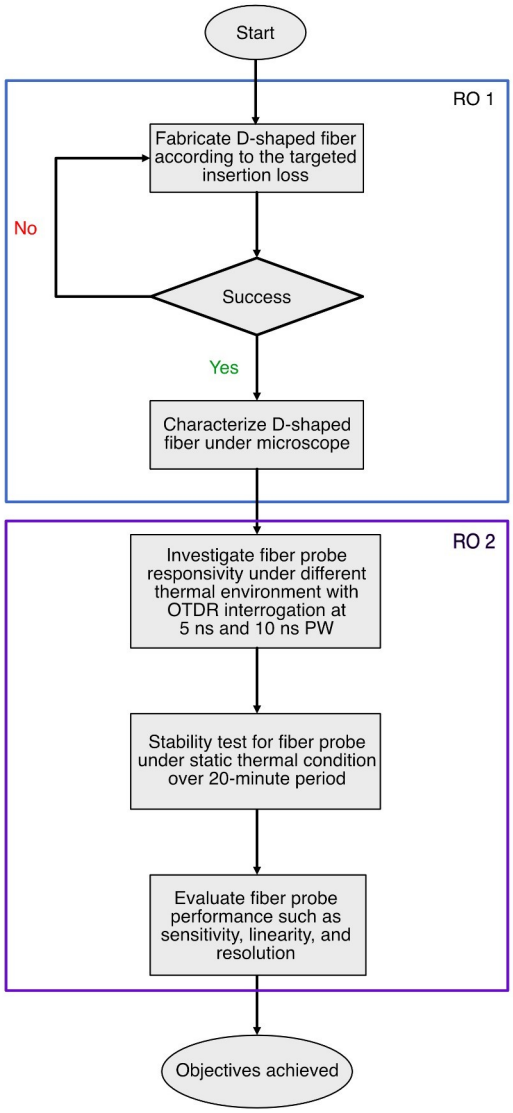


Fig. 3.1: Flowchart of the research methodology

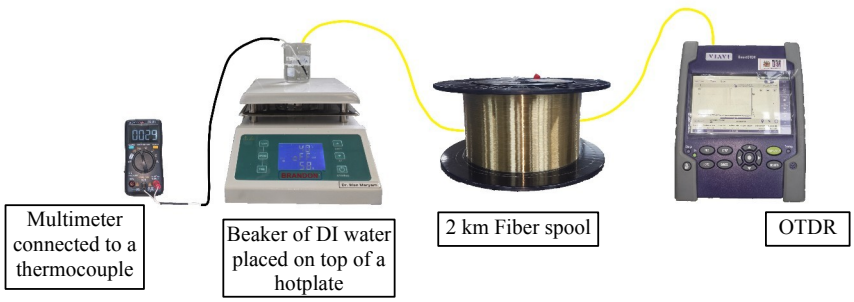


Fig. 3.2: Measurement setup for the characterization of the D-shaped fiber temperature sensor, including the OTDR, fiber spool, and reference thermocouple.

Throughout this experiment, VIAVI SmartOTDR was the main interrogation instrument. The OTDR was operated in standard single-ended reflectometry mode where the instrument launches a pulse into the optical fiber chain and records the time-resolved backscattered power retuning to the same port. Once plotted, the instrument produces a trace where the horizontal axis represents the distance along the fiber and the vertical axis represents the instantaneous backscattered power level in dB. Certain parameters, such as acquisition time, measurement range, and sampling resolution, were configured and held constant for each data acquisition run. Only PW is varied to 5 ns and 10 ns for each fiber configuration to investigate its effects to the trace resolution.

Furthermore, a 2 km pre-configured single-mode fiber spool with FC/UPC connector on both ends is connected to the OTDR port. This spool acts as a buffer between the D-shaped sensing region and the OTDR instrument dead zone, which is located immediately after the OTDR port. Any reflective event within this region is saturated by the Fresnel reflection and unresolved in the trace. The 2 km spool provide enough displacement margin for the sensing probe to be positioned on a clean, flat baseline region of the OTDR trace where reflective peak event can be resolved and measured.

The sensing element consists of SMF-28 fiber with a D-shaped cross-section that was fabricated by controlled mechanical side-polishing of the fiber cladding. One end of the probe is terminated with FC/UPC connector that is connected with the 2 km launch spool via an FC-FC adapter. The polished sensing region is located at the other end of the fiber and is submerged in a beaker of deionized water, which is positioned on top of the hotplate surface during the temperature measurement process. The far

end of the probe is terminated by an open-cleaved fiber face, which generates a strong Fresnel reflection at the end of the setup chain. This reflection appears as a sharp, high-amplitude peak on the OTDR trace, and it acts as a distance marker against which the D-shaped probe position can be referenced.

The primary measurand throughout this experiment is the OTDR reflection peak at the D-shaped sensing region (dB) relative to the applied temperature (℃). The independent variables that are varied throughout the study are the polishing depth of the D-shaped fiber (characterized by insertion loss at fabrication, ranging approximately from 2 dB to 8dB), the temperature applied to the sensing region, and the OTDR pulse width. An unpolished bare SMF-28 fiber was prepared, which functions as the baseline loss of unmodified SMF-28 in the measurement setup.

The remaining sections of this chapter describe the methodology used in this project with more detail. Section 3.2, layout the process of sensing elements fabrication process. Section 3.3 continues with description of the experimental setup and data acquisition process throughout this experiment.

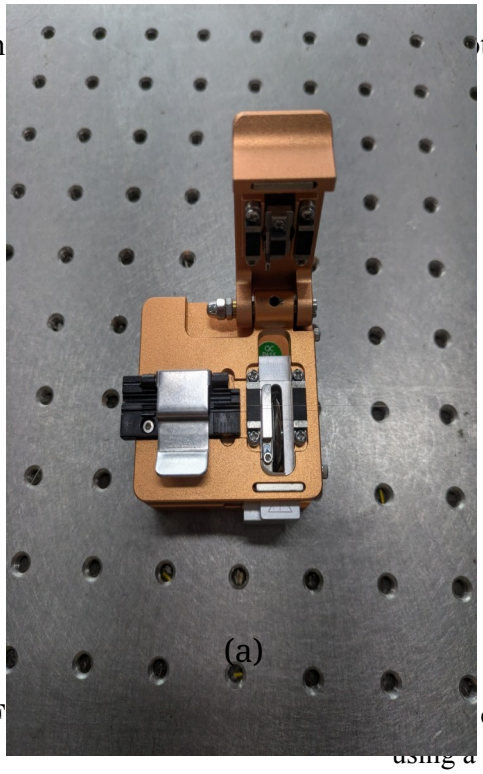
3.2 Fabrication of Sensing Probe

3.2.1 Fiber Preparation

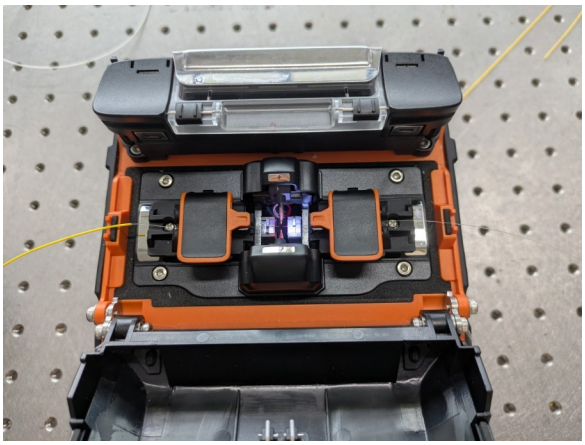
The fabrication of each sensing probe began with the preparation of the SMF-28 fiber ends that would be fusion-spliced to form the measurement chain described in Section 3.1. Proper fiber preparation is important to achieve low-loss, low-reflection splice joints. Any residual coating material, surface contamination, or end-face defect at the splice point introduces a discrete reflective or absorptive event on the OTDR trace within the 2 km launch spool junction. A poorly prepared splice

risks distorting the reflection peak at the sensing location. **Fig. 3.3 (a)** and **(b)** shows the step-by-step to fusion splice two optical fibers.

This process started with removal of the acrylate buffer coating from the terminal section of each fiber with a Signal Fire 9-in-1 mechanical fiber stripper. A single continuous strip of 30-40 mm was created, exposing the bare 125 µm silica cladding over this length. This strip length was long enough to provide the minimum bare fiber insertion length required by the Signal Fire AI-9 fusion splicer's V-groove and electrode geometry and also sufficient to accommodate the cleaver. Immediately after stripping, the exposed cladding surface was wiped with a lint-free wipe soaked with isopropyl alcohol (IPA) to remove the acrylate debris. Cleaning was performed by wiping the fiber in one direction rather than scrubbing it back and forth. This m t deposited back onto the cleaned surface.



(a)



(b)

F cleaving process and (b) the fusion splicing process using a Signal Fire AI-9.

Following that, the cleaned fiber was cleaved using the Signal Fire high-precision fiber cleaver to produce a flat, perpendicular end-face on the bare cladding. The cleaver was operated according to the standard procedure, with the fiber is

positioned onto the cleaver's fiber holder and the blade transverse through the cladding in a single controlled push. The angle of the cleaved fiber was assessed using the Signal Fire AI-9 fusion splicer's built-in inspection camera immediately prior to splicing. Fiber that have an end-face angle exceeding 1.0° or with a visible defects are rejected and, subsequently, re-cleaved. An angled or defective end-face at the splice produces two unwanted effects: it increases the geometric misalignment loss at the joint, and generates a non-specular Fresnel reflection that broadens and distorts the connector reflection event on the OTDR trace near the 2 km position.

Following the cleaving, the fiber was then permanently joined using fusion splicing to form the continuous optical path of the measurement chain. Fusion splicing was performed to establishes the mechanical and optical continuity between the D-shaped sensing element. A poorly executed splice could introduce a discrete reflection event on the OTDR trace other than reflection peak produced by the D-shaped sensing region itself.

Fusion splicing was performed using the Signal Fire AI-9 fusion splicer, as shown in **Fig.3.3 (b)**. The prepared fiber ends were loaded into the splicer's V-grooves with stripped and cleaved sections positioned across the electrode gap. To ensure a core-to-core alignment between both ends of the fibers before initiating the fusion, the splicer run an automated alignment routine using the built-in 3-axis motor. Once alignment was confirmed on the splicer display, the fusion arc was initiated. Simultaneously, the motor stages pushed the two fiber ends toward each other within a few micrometers to ensure no air gaps were entrapped at the joint. The total arc duration and electrode power were set by the splicer's pre-programmed SMF-to-SMF

fusion profile, with no modification applied throughout all splicing operations to ensure consistency across all probe samples.

Upon the arc completion, the splicer's camera system evaluated the joint and estimated the splice loss based on the geometric parameters on the fused region. A target splice loss of ≤ 0.02 dB was set as a criterion for all splices in the measurement chain. The splice loss contribution of ≤ 0.02 dB is negligible relative to the signal range and does not measurably alter the OTDR trace baseline against which the D-shaped reflection peak is measured. Splices estimated by the splicer to exceed this threshold were rejected and subsequently re-cleaved, and the splicing procedure was repeated until the acceptance criteria was met.

3.2.2 D-Shape Polishing

The D-shaped fiber probes were fabricated by controlled side-polishing of the spliced fiber using the custom setup as shown in **Fig. 3.4.** The objective of the polishing process was to remove cladding material from one side of the fiber until the evanescent field of the optical pulse was exposed at the polished surface. Five probe samples with different polishing depth, determined by the insertion loss monitored during the polishing process, were fabricated to investigate the relationship between polishing depth and temperature sensitivity.

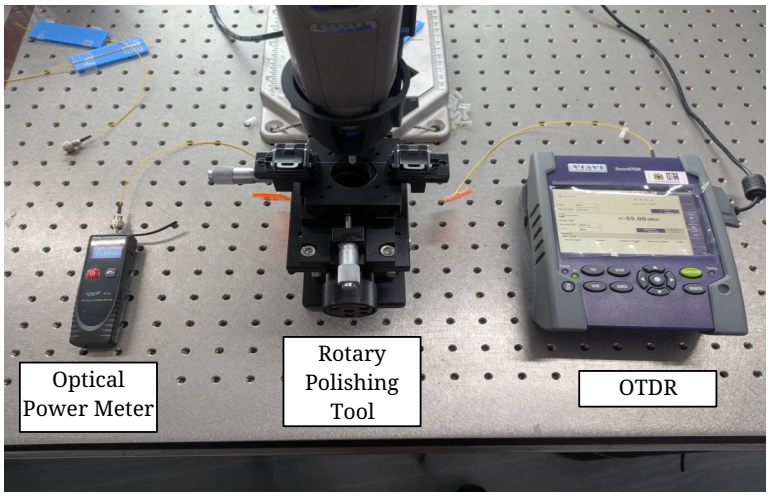


Fig. 3.4: Custom setup used to fabricate D-shaped fiber, featuring a rotary polishing tool and real-time power monitoring.

The rotary polishing tool used was a Dremel 200 series, fitted with a 4 mm diameter diamond-grit polishing bit and mounted vertically above the fiber on a rigid stand. The fiber under polishing was clamped horizontally in a multi-axis translation stage positioned beneath the bit. This stage provided independent control of the fiber's position relative to the bit by adjusting the contact between the bit and the fiber cladding with precision. The diamond-grit bit was selected in preference to conventional abrasive sandpaper for two reasons. First, the rigidity of the diamond bit result in a geometrically consistent flat polished surface compare to sandpaper which is more flexible and tends to produce a rounded cross-section that deviates from the ideal D-shaped geometry. Second, the diamond bit reduces fabrication time to under one minute per probe, compared to 15 minutes typically required with sandpaper-based polishing methods. The Dremel 200 rotational speed was varied during the polishing process with higher speeds during the initial rapid material removal of the outer cladding and lower speeds for a finer control of the final polishing depth as the process approached the target loss level. The specific speed settings during the

polishing process was controlled by the operator in response to the real-time optical power feedback described below.

The insertion loss during the polishing process was monitored continuously to determine when the target polishing depth has been reached. The VIAVI SmartOTDR was configured in continuous-wave (CW) visual fault locator (VFL) mode and connected to the fiber. The Wintact WT65 optical power meter was connected to another end of the fiber to measure the transmitted power in real time. As the cladding material was removed progressively, the increasing evanescent exposure at the polished surface increased the insertion loss, measurable as a decreasing transmitted power reading on the power meter. Polishing was stopped when the transmitted power had dropped by the target loss amount relative to the initial power measured before the start of each fabrication run. The five target loss levels used in this work are approximately 2 dB, 4 dB, 6dB, 7dB, and 8dB, each represent a progressively deeper polishing depths.

3.2.3 Geometric Characterization of the Sensing Probes

Following fabrication, each D-shaped probe was examined under an optical microscope to verify that the polishing process had produced the intended geometric modification of the fiber cross-section and to quantify the residual cladding thickness at the polished face. This geometric characterization not only help with providing proof of the polishing depth inferred from the insertion loss monitoring during the fabrication but also the physical length of the fiber which is required to estimate the evanescent fraction η.

Each polished probe was placed on the Motic BA310 microscope stage with the polished face oriented horizontally. This side-view imaging geometry produces

and image in which the full width of the fiber cladding and polished edge is visible. The microscopes images confirmed that the polishing process produced a flat surface on one side of each fiber, consistent with the D-shaped cross-sectional geometry described in Section 2.4.1.

3.3 Experimental Setup for Temperature Sensing

3.3.1 OTDR Configuration and Parameter Selection

In this experimental setup, the OTDR is connected to the 2 km launch spool via the SC/APC port. The far end of the spool connected to the D-shaped sensing probe with FC/UPC connector with each of the components connected via an FC/FC adapter. The far end of the probe was open-cleaved terminated. This series of arrangement placed the D-shaped sensing region at a distance of approximately 2 km plus the length of the input connector at the OTDR port and the probe.

Every OTDR measurement begins with a strong Fresnel reflection at the instrument's output connector, which caused by the refractive index discontinuity at the glass-air-glass interface of the mated SC/APC connector. This reflection temporarily saturates the OTDR receiver, making the detector unable to resolve any signal for a short distance immediately after the connector. This short region is known as the instrument dead zone. The 2 km launch spool displaced the sensing region far beyond this dead zone, ensuring that the reflection peak generated by the sensing probe sits on an entirely undisturbed, flat baseline region of the trace. Without this displacement, the sensing event would fall within the saturated dead zone and could not be resolved.

Beyond functioning as dead-zone clearance, the launch spool also provides a well-characterized linear backscatter slope over 2 km of standard SMF-28 prior to the sensing region. This slope provides a stable reference baseline against which the reflection peak at the D-shaped probe location is measured. Any deviation from this baseline slope in the vicinity of the probe can be attributed to the sensing element rather than caused by the instrument itself.

All OTDR acquisition parameters were configured at the start of each experimental runs and held constant throughout the process to ensure that variations in the OTDR trace are attributable solely due to the temperature change on the sensing probe. **Table 3.1** summarizes the selected parameters and their rationale.

Table 3.1: OTDR Measurement Parameters

Parameters	Setting
Operating Wavelength	1550 nm
Measurement Range	5 km
Acquisition Time	10 s
Sampling Resolution	32 cm
Pulse Width (PW)	5, 10 ns

The 1550 nm wavelength was selected for two reason. First, as stated in Section 2.3.2, the evanescent penetration depth into the external medium at the polished D-shaped surface increases with wavelength. At 1550 nm, the penetration depth is approximately 400 nm in aqueous media, compared to approximately 280 nm at 1310 nm. This 43% increase enhances the sensor's sensitivity to changes in the medium's optical properties. Second, the maximum attenuation of SMF-28 at 1550 nm is approximately 0.18 dB/km, compared to approximately 0.32 db/km at 1310 nm [32]. Over the 2 km launch spool, the difference between both wavelength is about

0.32 dB of additional optical loss. For a system that relied on a minute change in value, the difference is non-trivial.

The measurement range and acquisition time were 5 km and 10 seconds, respectively. Measurement range is the maximum distance that the OTDR will monitor and display in the trace. Said range is more than enough to accommodate the full length of the measurement chain with sufficient margin to include the Fresnel reflection occurred at the cleaved end of the fiber. The acquisition times, however, is the duration for which the OTDR sends out laser pulses and collects the returning backscattered light to be averaged into a single trace. While longer acquisition time theoretically improves the SNR by suppressing random noise, a 10-second duration provides a viable trace without unnecessarily prolonging the data collection process. Acquisition times shorter than 10 seconds produces visibly noisier trace while acquisition times longer than that provides no visible improvement in trace quality. The sampling resolution was set to ensure the highest density of data points. For 5 km measurement range in VIAVI SmartOTDR, it was 32 cm. Selecting the finest available resolution ensures that the localized attenuation changes within the sensing region are captured with high precision.

Variation in pulse width during the measurement process was done to identify the spatial resolution and SNR in OTDR measurement. PW determines the physical length of the fiber at which the pulse occupies at a given moment. The higher the pulse length, the higher its optical power, but the less resolved the sensing points become. For every temperature measurement across all probe samples and all temperature steps, both 5 ns and 10 ns PW were used. This variation helps with

identifying the optimal PW to be used by evaluating the trade-off between spatial resolution and SNR of both values.

3.3.2 Heating Element and Thermal Control

In this experiment, Brandon BHS-II-4D Digital Ceramic Hotplate Stirrer was used as the heating element during the data acquisition process. This hotplate has a flat ceramic heating surface with a digital built-in surface temperature sensor that provides a closed-loop feedback to the hotplate's internal controller to maintain surface at the set temperature once thermal equilibrium is reached. The D-shaped sensing region of each probe was submerged in a beaker of deionized water placed on the hotplate surface. Deionized water was selected as the thermal medium due to its high thermal conductivity relative to air, ensuring uniform temperature distribution around the sensing region. The beaker was positioned at the center of the hotplate ceramic surface where the temperature distribution is the most uniform.

The temperature of the deionized water at the sensing region was measured using a thermocouple probe connected to a multimeter with temperature measurement capability. This approach eliminates reliance on the hotplate's built-in digital display, which reflects the ceramic surface temperature rather than the actual temperature of the water medium surrounding the fiber. The thermocouple measurement therefore provides a more accurate and direct reading of the thermal condition experienced by the sensing region.

3.3.3 Data Acquisition Process

Each of the six probe samples was characterized in a separate measurement sessions. Before the start of every session, the full measurement chain was assembled

from scratch to ensure that no residual thermal from previous session affected the results. The unpolished control probe was measured to act as a baseline temperature-dependent optical response against which the other five D-shaped probes will be compared.

At the start of each session, all FC/UPC connectors and SC/APC instrument port connector were cleaned using a one-click optical fiber cleaner. Connector contamination is the most common source of random reflection events on OTDR traces. Following the connector cleaning and chain assembly, a smart test was run to confirm the absence of anomalous reflection or loss events attributable to contaminations or misalignment in any connector junction. Only after this condition was satisfied the measurement session can be run.

The probe was submerged in the beaker of deionized (DI) water before the hotplate was turned on to ensure the operator's safety. The thermocouple tip was positioned alongside the sensing region inside the beaker to monitor the water temperature directly. The initial temperature set point of the hotplate was set to 40 ℃. If a temperature below that was used as a set point, the measurement session will take a longer time or a more direct cooling approach is needed for the water to reach said temperature. It is easier and less time-consuming to heat the water to temperature slightly above the room temperature than the former approach. Once the thermocouple reading had stabilized at 40 ℃, a 2-minute stabilization period was taken before the first OTDR trace was acquired for the 5 ns PW. This stabilization period allows for the fiber to reach thermal equilibrium with the surrounding water, which is essential to prevent a measurement lag where the OTDR records a trace at a temperature that the fiber has not yet reached.

Following acquisition of the 40 ℃ trace, the hotplate set-point was incremented until the thermocouple reading reached the next target temperature of 45 ℃. The same 2-minute stabilization period was taken at each subsequent step before trace acquisition. This procedure was repeated at 45℃, 50℃, 55℃, 60℃, 65℃, 70℃, 75℃, 80℃. After first session was done, the water was let to return to 40 ℃ for the next measurement session for 10 ns PW. Since the experiment involve unavoidable noise, the common point-to-point calculation might not fully capture accurate sensitivity of the sensing probe. Hence a more accurate Ordinary Least Square (OLS) method is used to calculate the sensitivity:

$$S=\frac{\sum_{n}^{i=1}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{n}^{i=1}(x_i-\bar{x})^2}, \tag{16}$$

where x_i is the temperature at point i, $\bar{x}$ is the average temperature of all points, y_i is the loss value at point i, and $\bar{y}$ is the average loss of all points.

A stability test was conducted on the bare fiber control probe and the D-shaped probe with the most consistent linearity to evaluate the repeatability and consistency of the OTDR measurement system under static thermal conditions. Both probes were submerged in the beaker of deionized water and held at a fixed temperature of 60 ℃. At this condition, the OTDR traces were acquired 10 consecutive times at 2-minute intervals without any change to the temperature setting of the measurement chain configuration. The acquired traces were then compared to assess the degree of variation between repeated measurements at a constant temperature. A low variance across the 10 acquisitions would confirm that the measurement system is stable and

that any temperature-dependent changes observed during the main characterization process can be attributed to the thermal stimulus rather than random noise.

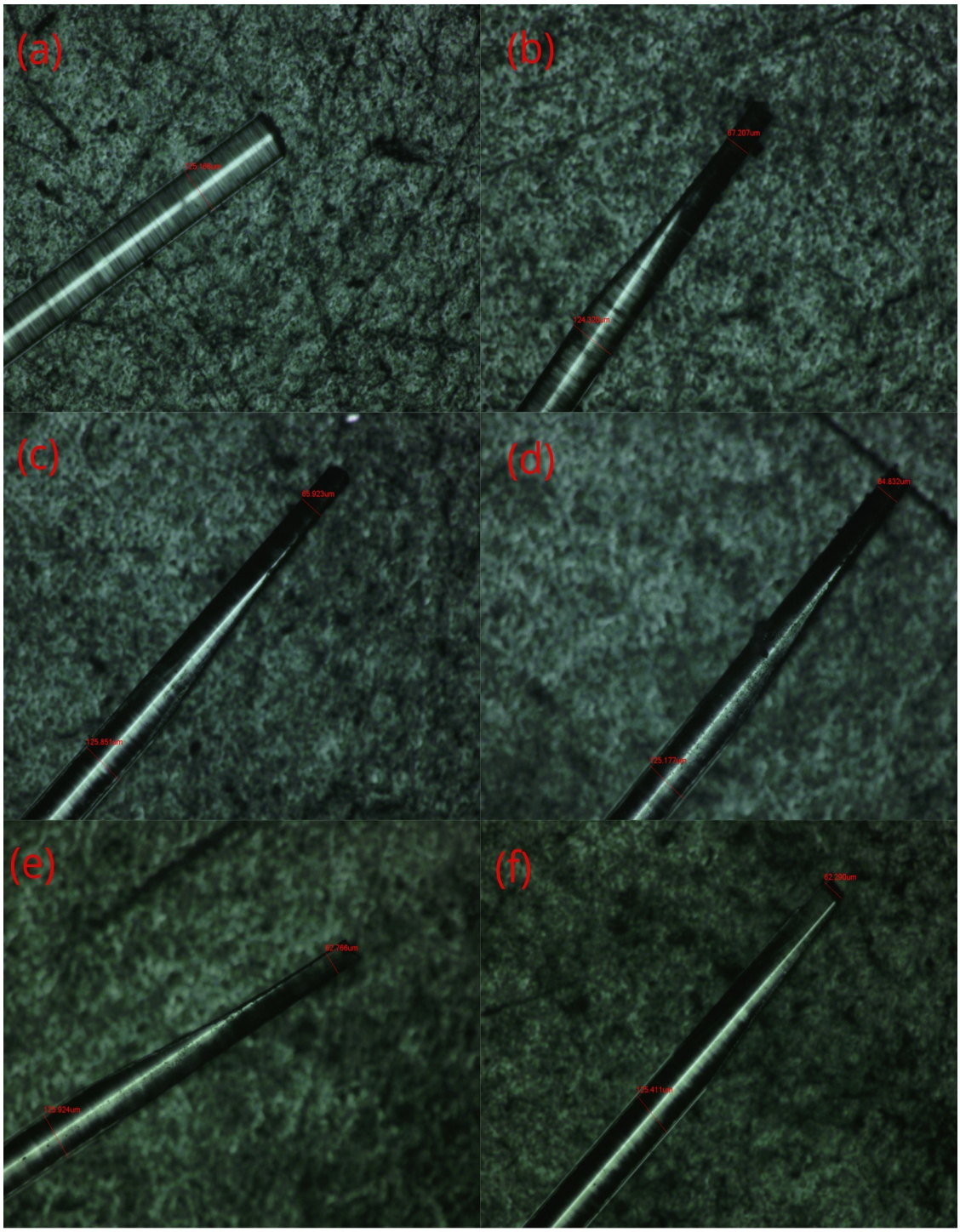
Chapter 4: Results and Discussion

4.1 Sensing Probes Geometric Characterization

Following the side-polishing process, each of the five D-shaped fiber probes was subjected to optical microscope inspection. This characterization serves to visually verify that the mechanical side polishing successfully altered the cross-section geometry from a symmetric cylinder to an asymmetric D-shape.

To evaluate the visual boundaries, digital line-profiling was conducted within the Toupview imaging software application. These digital lines serve as a visual approximation of the progressive decrease in cladding thickness across the sample set, rather than as an absolute geometric measurement. Crucially, within optical fiber sensing architectures, the recorded insertion loss (dB) acts as a better and more accurate indicator of evanescent field exposure than physically measured diameter. This method align with works by other researchers including Jafry, Tarmizi and colleagues [10], [53].

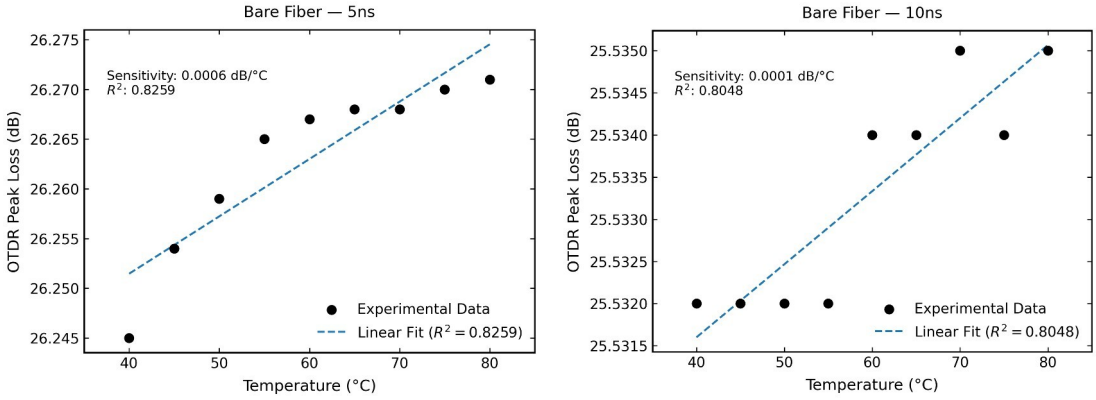
Fig. 4.1 presents the side-view microscope images collected for the six probe configurations. A consistent progression in material removal is visible from the shallowest to the deepest polishing depth. The captured images confirm that the diamond-grit polishing bit successfully generated a uniform, flat surface parallel to the longitudinal fiber core axis across the entire imaged sensing length.



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system, an unpolished standard single-mode fiber was first characterized. This

baseline assessment is critical to verify that any subsequent sensitivity enhancement observed in geometrically altered probes originates from the intentional exposure of the evanescent field. The bare fiber was subjected to identical thermal cycle ranging from 40 ℃ to 80 ℃ with 5 ℃ increments. **Fig. 4.2** illustrates the recorded optical response, plotting the OTDR peak loss as a function of temperature under two distinct pulse widths of 5 ns and 10 ns.



(a) (b)

Fig. 4.2: OTDR peak loss as a function of temperature for the reference bare fiber at (a) 5 ns, and (b) 10 ns interrogation pulse widths

The experimental data for both interrogation parameters reveals a consistent and stable baseline response across the entire thermal span. This steady progression demonstrates the stability and reliability of the OTDR interrogation system under temperature variation. When interrogated with a 5 ns pulse width, the reference fiber exhibits a baseline peak loss of 26.245 dB at 40℃ and gradually increase to 26.271 dB at 80℃. This shows a positive temperature sensitivity of 0.00006 ℃dB/with a moderate linear correlation coefficient (R^2) of 0.8259. Similarly, under a 10 ns pulse width interrogation, the baseline peak loss hover around 25.53 dB across the entire temperature span. The resulting linear regression yields an even lower sensitivity of 0.0001 dB/℃ with R^2 of 0.8048.

The physical mechanism underlying this response is directly tied to the waveguide geometry of standard SMF-28 fiber as explained in Section 2.4. The core is shielded by a thick 125 μm silica cladding layer, which completely isolates the guided optical power from the external environment. Since the surrounding medium cannot interact with the optical fields, the external refractive index variations of the water bath during heating have negligible impact on the core guidance conditions. Consequently, the microscopic fluctuations observed in the bare fiber traces are entirely governed by the intrinsic properties of the silica glass itself, specifically the thermo-optic coefficient ($\partial n/\partial T$) and the coefficient of thermal expansion (a) of the germanium-doped core [46]. These intrinsic material properties slightly alter the backscattering coefficient and the localized attenuation of the silica grid, manifesting as the minute fractional decibel variations captured in the plots.

Furthermore, comparing the two curves highlights how the choice of pulse width subtly impacts data interpretation even in a baseline setup. While both pulse widths successfully map the expected temperature profile, the 5 ns pulse width data traces a more coherent linear progression. The 10 ns pulse width data displays a staggered, stepped behavior, such as clumping identically at 25.532 dB from 40 ℃ to 55℃ before stepping up to 25.534 dB. This behavior indicates that the wider 10 ns pulse introduces a higher spatial interrogation length, which averages minor local backscatter variations and limits the resolution of extremely, fine sub-millidecibel signal changes. Ultimately, these control results confirm that an unmodified fiber is structurally less suitable for sensitive thermal monitoring.

4.3 Temperature Sensing Performance of D-Shaped Fiber

The experimental data reveals that the temperature sensitivity of the sensor is determined by the initial insertion loss of the probe, which serves as an operational proxy for the proximity of the flat polished boundary to the fiber core. **Fig. 4.3** illustrates the combined linear regression curves and corresponding experimental data points for all five D-shaped fiber configurations under an optimized 5 ns interrogation pulse width.

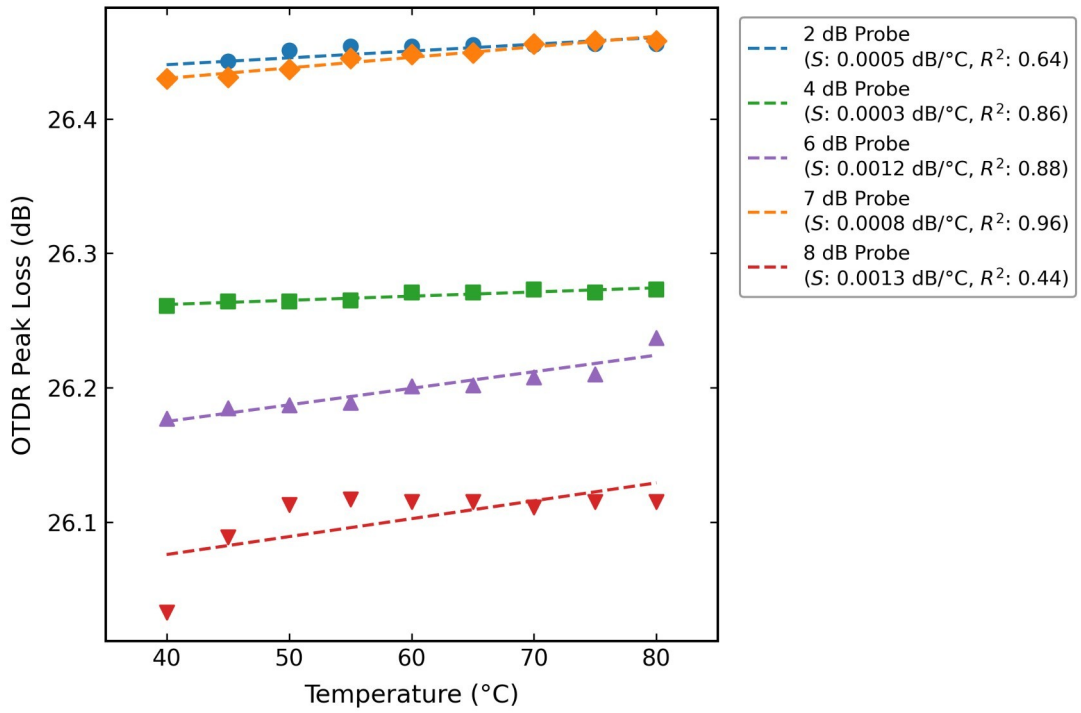


Fig. 4.3: Combined temperature response curves and linear regression fits for the 2 dB, 4 dB, 6 dB, 7 dB, and 8 dB D-shaped fiber probes interrogated at a 5 ns pulse width.

At a shallow polishing depth, represented by the 2 dB probe, the sensor exhibits a low sensitivity of 0.0005 dB/℃ with a poor linear correlation of 0.6394. As the polishing depth increases, a distinct transition in performance occurs. The 4 dB probe demonstrates a stable but moderate response of 0.0003 dB/ ℃ sensitivity and 0.8637 R^2. A surge in performance is observed when the insertion loss reaches 6 dB and 7 dB. The 6 dB probe shows increase in sensitivity to 0.0012 dB/℃ (R^2=0.8843).

The 7 dB probe achieves the optimal balance among the linear configurations, yielding a sensitivity of 0.0008 dB/°C with an exceptional linear correlation of $R^2 = 0.9552$. This strong linearity is highly desirable for practical sensing applications as it simplifies calibration and digital signal processing requirements.

The increase in sensitivity and linearity across different probe configurations can be attributed to the geometric effect of the D-shaped polishing on the evanescent field exposure. As the insertion loss increases across fabricated probes, the polishing depth increases accordingly, bringing the fiber core progressively closer to the external medium. This geometrically increases the fractional power of the evanescent field η available for interaction with the surrounding medium, as described in Section 2.5.2. It is important to note that this relationship describes a comparison across probes of different fixed polishing depths, and is distinct from the temperature-dependent behavior of η within a single probe, which is governed by a different mechanism discussed below.

From **Fig. 4.4**, it can be seen that as the temperature increase, the peak loss follows suit. To understand this trend, the underlying optical mechanisms must be considered. While the sensor utilizes a D-shaped geometry to expose the fiber's evanescent field to the external DI water, evanescent field absorption framework does not fully account for the observed signal modulation. This is because primary signal captured by the OTDR is not a distributed transmission attenuation along the polished zone, but rather a discrete event terminated by the open-cleaved fiber endface.

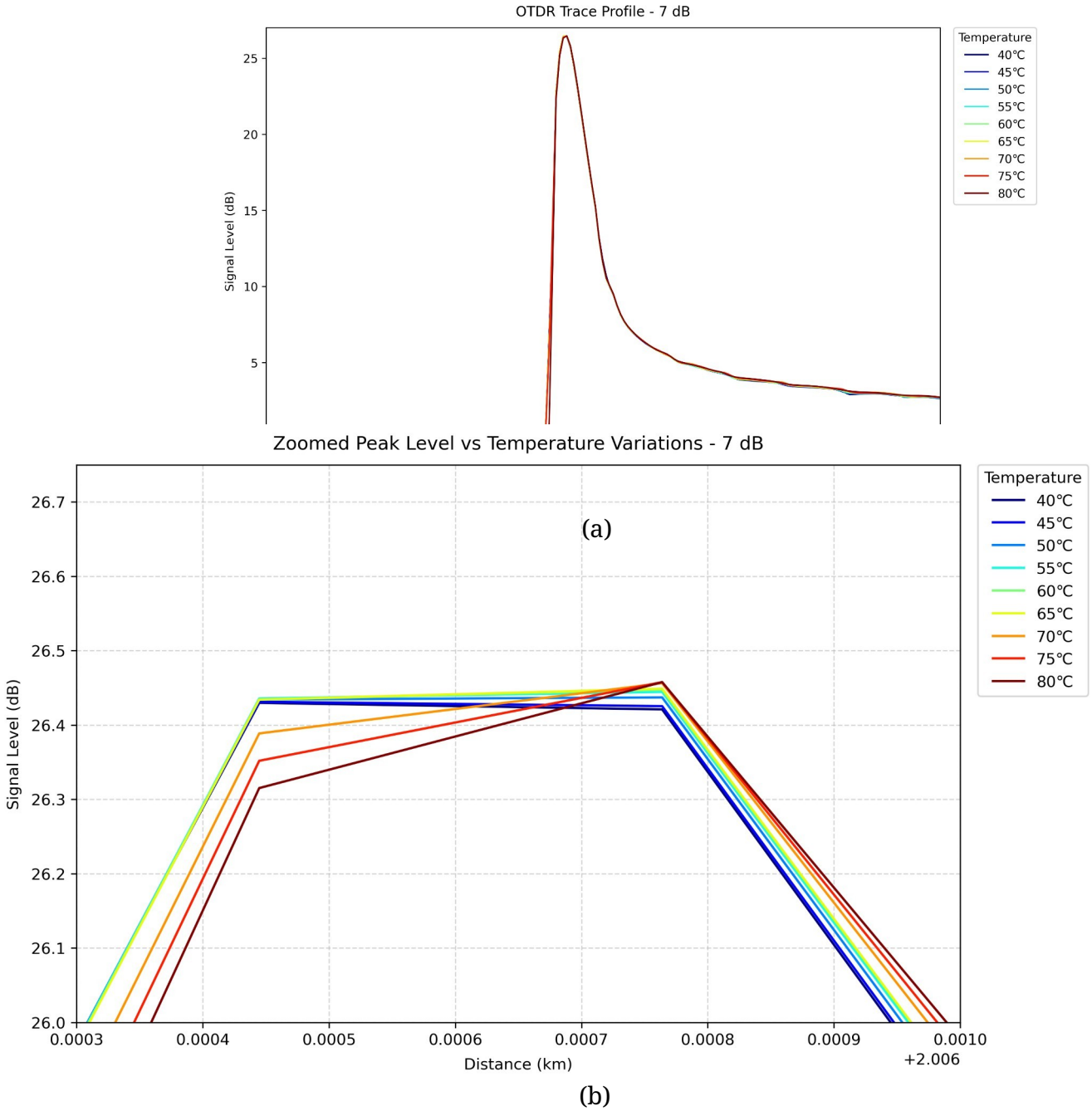


Fig. 4.4: OTDR trace at (a) the 2 km marker and (b) zoomed peak level against temperature variations for the 7 dB D-shaped sensing probe across a temperature range of 40℃ to 80℃ at 5 ns pulse width.

At this sharp, glass-to-water interface, the signal is primarily governed by the Fresnel reflection mechanism outlined in Equation 5. DI water possesses a strong negative thermo-optic coefficient ($\partial n/\partial T \approx -1\times10^{-4} K^{-1}$). As the water temperature rises from 40 ℃ to 80 ℃, its refractive index decreases, hence widening the refractive index contrast between the silica fiber core and the surrounding medium. As a result,

the magnitude of the reflection *R* simultaneously increase, forcing more optical power back toward the OTDR detector, which registers as an increasing peak level on the trace.

The necessity of the D-shaped geometry within this Fresnel-based sensing mechanism is confirmed by the bare fiber control results in Section 4.2, which demonstrated lower linearity and sensitivity despite possessing an identical cleaved end-face submerged in the same medium. This indicates that the end-face Fresnel reflectance alone is insufficient to produce a measurable temperature response without the incorporation of D-shaped fiber on the sensing region. The D-shaped geometry ensures that the guided mode interacts with the surrounding water along the sensing zone, making the effective optical boundary condition at the end-face responsive to the temperature-dependent refractive index of the water. However, present waveguide literature lacks a unified empirical formula that reconciles the simultaneous effects of evanescent field exposure and Fresnel reflection.

While sensitivity trends upward with deeper polishing, the experimental data reveals a physical limit to this phenomenon. As shown in **Fig. 4.3**, the 8dB probe registers the highest sensitivity slope (0.0013 dB/ ℃), but with degraded R^2 of 0.4422. The result of the 8 dB probe might suggest that the fiber has been overpolished, hence, exposing the fiber core. When the core is exposed to the environment, the guided light might interacting with the environment directly instead of relying on the evanescent field modulation [54]. Therefore, while the 8 dB displays the highest sensitivity, its high non-linearity makes it unsuitable for reliable calibration, indicating the 7 dB configuration as the optimal threshold for maximizing evanescent interaction.

A critical finding of this study is the dependency of point-sensing resolution on the spatial parameters of the interrogation instrument. **Table 4.1** shows the comparative linear regression metrics across both 5 ns and 10 ns pulse width setting for all fabricated probes.

Table 4.1: Comparative sensitivity and linearity across interrogation pulse widths

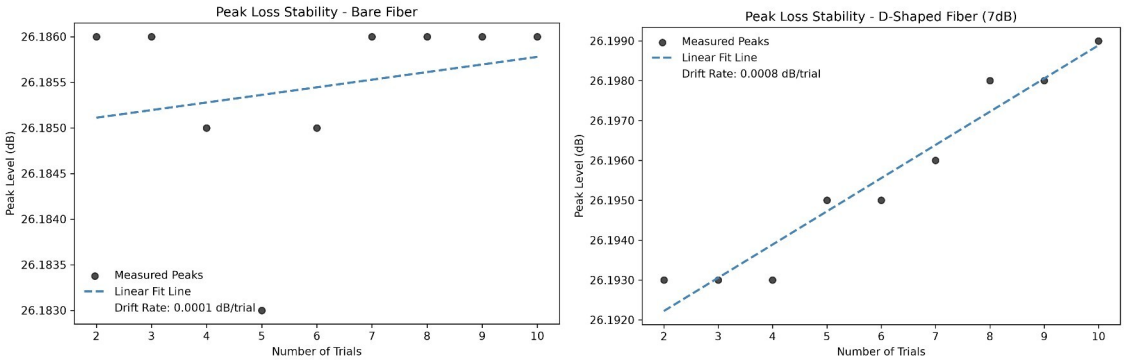
Probe Configuration	5 ns Pulse Interrogation		10 ns Pulse Interrogation	
	Sensitivity (dB/℃)	Linearity (R^2)	Sensitivity (dB/℃)	Linearity (R^2)
2 dB	0.0005	0.6394	0.0001	0.7408
4 dB	0.0003	0.8637	0.0003	0.8578
6 dB	0.0012	0.8843	0.0000	0.7500
7 dB	0.0008	0.9552	0.0001	0.7147
8 dB	0.0013	0.4422	0.0000	0.2182

The data shows that migrating from a 5 ns pulse to a 10 ns pulse flattens the recorded temperature response. For the highly responsive 6 dB and 7 dB probes, the measurable sensitivity drops to 0.0000 dB/℃ and 0.0001 dB/℃ respectively, while the R^2 parameters diminish. This behavior is due to the decrease in spatial resolution, which is related to the PW. A 10 ns PW occupies a physical event of approximately 1 meter along the fiber length. By reducing the interrogation parameter to a 5 ns pulse width, the occupied spatial window is roughly 0.5 m, suitable with the point sensor that extend across only a few millimeters.

4.4 Thermal Stability of Sensing Probes

To validate the reliability and repeatability of the interrogation system, a thermal stability analysis was conducted under static environmental conditions. As detailed in the methodology section, both the unpolished reference bare fiber and 7dB

D-shaped sensing probe were submerged in a deionized water bath held at constant temperature of 60℃ with a series of 10 consecutive OTDR traces were acquired at 2-minute intervals over a total duration of 20 minutes. This test is critical to demonstrate that the temperature-dependent optical adjustments recorded during the primary characterization phase are driven strictly by external thermal condition rather than random environmental noise, instrumentations fluctuations or localized measurement lag. **Fig. 4.5** illustrates the recorded optical loss for stability tests of bare fiber and 7 dB D-shaped fiber probe.



(a) (b)

Fig. 4.5: Peak signal level of stability test over ten consecutive trials at a static temperature of 60°C over 20-minute period for (a) the reference bare fiber control probe and (b) 7 dB D-shaped sensing probe.

The statistical parameters extracted from the 10 stability trials for both probe configurations are compiled in **Table 4.2.**

Table 4.2: Statistical summary of peak signal level stability over 10 consecutive trials at 60℃.

Parameter	Bare Fiber	7 dB D-Shaped Fiber
Mean Peak Loss (dB)	26.18550	26.19530
Standard Deviation (dB)	0.00097	0.00236
Maximum Peak Value (dB)	26.18600	26.19900
Minimum Peak Value (dB)	26.18300	26.19300
Peak-to-Peak fluctuation (dB)	0.00300	0.00600
Linear Drift Rate (dB/trial)	0.00010	0.00080

As shown in **Fig. 4.5,** The stability profile for the reference bare fiber demonstrates the intrinsic baseline performance of the interrogation loop. Over the course of the 20 minutes monitoring period, the bare fiber exhibited an exceptionally stable response, maintaining an identical peak loss value of 26.186 dB across 7 of the10 trials. The total peak-to-peak fluctuation range was confined to a mere 0.003 dB, yielding an extremely low standard deviation of 0.00097 dB. When subjected to linear regression, the bare fiber data reveals a negligible slope of 0.0001 dB/trial. This lack of linear correlation indicates that the minuscule variations observed correspond to the random thermal noise floor of the OTDR instrument receiver and slight optical power fluctuations in the internal laser source. This confirms that the mechanical splice joints, and launch spools, and physical connector interfaces remain structurally stable and do not introduce systematic drift into the measurement loop.

For the 7 dB D-shaped probe, the stability performance reflects the structural modification made to the fiber geometry. The probe recorded a mean peak loss of 26.19530 dB with a total fluctuations range of 0.006 dB between the minimum (26.193 dB) and maximum (26.199 dB) values. This minor variation translates to an acceptable standard deviation of 0.00236 dB. Unlike unpolished control fiber, the 7 dB probe displayed a subtle, highly linear upward drift of 0.0008 dB/trial. This shift is not indicative of sensor instability rather result of exposing the evanescent field directly to the surrounding water medium. This structural change results in higher sensitivity to external environment than the standard bare fiber. As a result, while the bare fiber remains completely irresponsive to the micro-change in the environment, the D-shaped probe registers minuscule localized variations and settling trends within the experimental setup.

Chapter 5: Conclusion

5.1 Summary

This thesis successfully demonstrated the development and experimental characterization of a point temperature sensor that integrates a D-shaped single-mode optical fiber with a commercial Optical Time-Domain Reflectometer (OTDR). By monitoring the amplitude fluctuations of a localized reflection peak, the sensing architecture effectively tracked ambient temperature variations across a thermal range of 40°C to 80°C. The experimental results revealed that the sensor's temperature sensitivity and linearity are heavily governed by its initial insertion loss, which serves as an operational proxy for cladding reduction. Among the fabricated configurations, 7dB probe was identified as the optimal design, delivering a stable sensitivity of 0.0008 dB/°C and an exceptional linear correlation coefficient (R^2) of 0.9552. Conversely, while a deeper 8 dB probe registered a higher sensitivity slope of 0.0013 dB/°C, its linearity severely degraded to an R^2 of 0.4422. This non-linear breakdown points to a structural threshold where directly breaching the core matrix induce chaotic modal boundary discontinuities rather than uniform, stable evanescent field modulations.

Furthermore, the point-sensing performance demonstrated a strict dependency on the spatial parameters of the interrogation instrument. 5 ns pulse width emerged as an absolute operational requirement to resolve the localized, millimeter-scale sensing zone. When the interrogation parameter was migrated to a wider 10 ns pulse width, the recorded temperature response completely flattened to approximately 0.0000 dB/°C. This degradation occurs because the wider 10 ns pulse occupies a physical window of roughly 1 meter, causing the OTDR mathematically average the highly

localized transmission loss modulation against the vast background signal of the surrounding medium, thereby effectively masking the thermal perturbations.

Finally, thermal stability analysis conducted at a static temperature of 60°C over a 20 minutes period validated the high reliability of the setup. The unpolished fiber control probe exhibited a minuscule standard deviation of 0.00097 dB, while the 7 dB D-shaped probe demonstrated an acceptable standard deviation of 0.00236 dB with a subtle linear drift of 0.0008 dB/trial. Rather than indicating sensor instability, this minor shift successfully captured real-time micro-thermal settling behavior and slow thermal equilibration trends within the liquid medium as the hotplate system fine-tunes

5.2 Future Work and Recommendations

This study demonstrates the possibility of developing a structurally simple temperature OFS. In particular, this system is suitable to be applied in a field work environment that extend over a long range. This architecture is also appropriate in an extreme settings such as on the oil and gas pipelines, power lines, and deep sea infrastructure.

To build upon the foundational sensor system established in this study and translate it into a practical tool for industrial or environmental monitoring, several approaches are recommended for future research. First, future work should focus on the incorporation of thermally responsive coatings directly onto the flat polished surface of the fiber probe to significantly amplify intensity modulation. Examples of coating that is worth exploring are Zinc Oxide (ZnO), Molybdenum Disulfide (MoS_2) and polydimenthysiloxane (PDMS) coating which represents high potential for this

application as reported in other literature [5], [8], [26]. Since unmodified silica possesses a low intrinsic thermo-optic coefficient, leveraging such coatings will potentially yield a drastically superior thermal responsivity compared to the bare configurations.

Additionally, the system should be expanded to explore multi-point and quasi-distributed scaling by utilizing the distance-resolved loss localization capabilities inherent to conventional OTDR instruments. Cascading multiple optimized D-shaped sensing probes in series along a single strand of fiber would allow researchers to evaluate the architecture's capacity for parallel, real-time temperature field monitoring.

Lastly, future work should prioritize the development of mathematical model that unify evanescent field modulation and Fresnel reflection. Currently, there is a lack of analytical equations or theories to reconcile how lateral cladding remove profile explicitly enhanced the reflection amplitude. Developing an exact boundary-value electromagnetic model would provide deeper insight into the physics of this architecture and allow for the precise geometric optimization of future intensity-based point sensors.

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